

Speech Processing

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Review of Signal Processing: LTI Systems

- A **Linear Time-Invariant (LTI)** system is completely characterized by its **impulse response** = response of the system to $\delta[n]$.
- Response of LTI system for any arbitrary input is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

- Discrete time Fourier Transform of a signal is given as $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

$$\omega = 2\pi \frac{F}{F_s} \quad -\pi \leq \omega \leq \pi$$

- ω in rad/sample; $\omega=0$ to π is considered as general frequency range (ignoring negative values $-\pi$ to 0).
- F_s : sampling rate in samples/sec or Hz
- F : continuous time frequency in cycles/sec or Hz

Frequency response

The **frequency response of LTI system** is the DTFT of the impulse response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

$|H(e^{j\omega})|$ (Magnitude response)
 $\angle H(e^{j\omega})$ (Phase response)

Input-output relation: $y[n] = x[n] * h[n]$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

DTFT properties

Property	Time-Domain Operation	DTFT Relation
1. Delay (Time Shift)	$y[n] = x[n - n_0]$	$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$
2. Convolution	$y[n] = x[n] * h[n]$	$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$
3. Symmetry (Time Reversal / Conjugation)	$y[n] = x[-n]$	$Y(e^{j\omega}) = X(e^{-j\omega})$
	$y[n] = x^*[n]$	$Y(e^{j\omega}) = X^*(e^{-j\omega})$

System function

System function $H(z)$ standard form

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

- **Zeros**: roots of the numerator
- **Poles**: roots of the denominator
- Poles and zeros are plotted in complex z -plane. $z = r^* e^{j\omega}$
- $|z|=r=1$ is called **unit circle** in z -plane

Frequency response from $H(z)$

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}}$$

- **Zeros near unit circle** → spectral **nulls (valleys)**
- **Poles near unit circle** → spectral **peaks (resonances)**

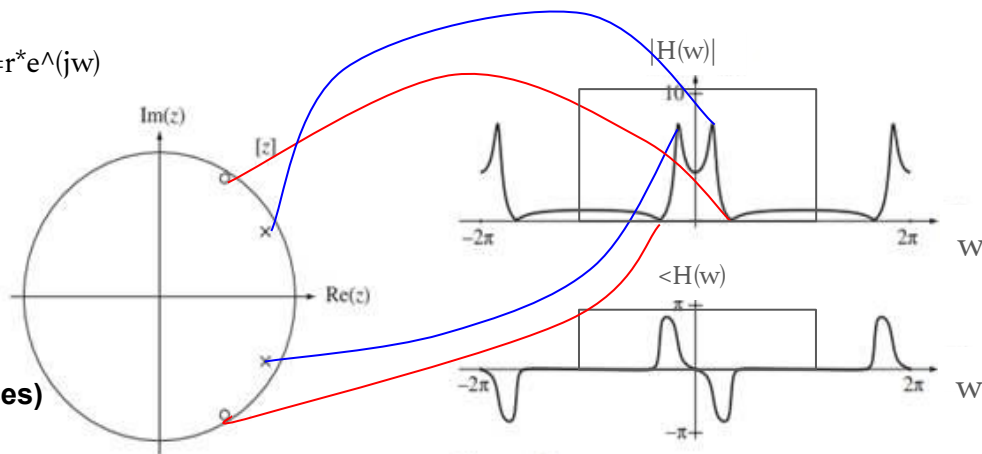


Figure 9.16

Pole-zero diagram of the transfer function,

$$H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Figure 9.17

Magnitude and phase frequency response of the system whose transfer

$$\text{function is } H(z) = \frac{z^2 - 0.96z + 0.9028}{z^2 - 1.56z + 0.8109}$$

Autocorrelation of a signal

Auto correlation is similarity of a signal with its delayed-self.

Its a function of delay: $R_{xx}[k] = \sum_{n=-\infty}^{\infty} x[n] x^*[n - k]$

$R_{xx}[0] = \sum_{n=-\infty}^{\infty} |x[n]|^2$ gives energy of the signal.

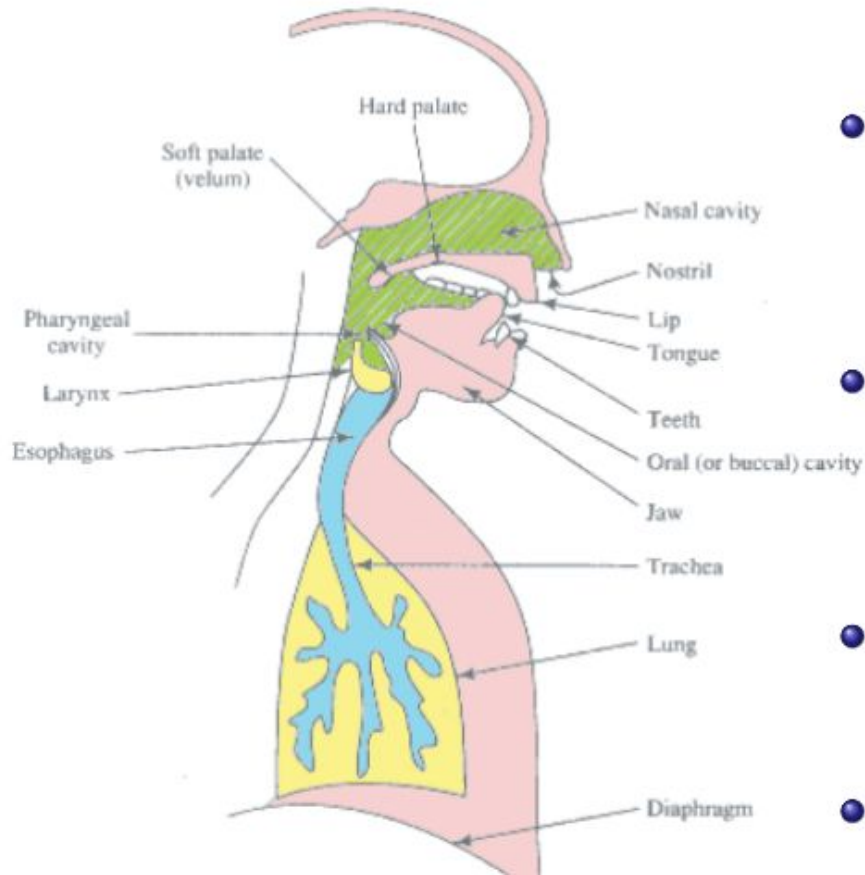
Autocorrelation is even symmetric function: $R[-k]=R[k]$.

Autocorrelation of a periodic signal is also periodic with same period.

DTFT of autocorrelation gives power spectral density.

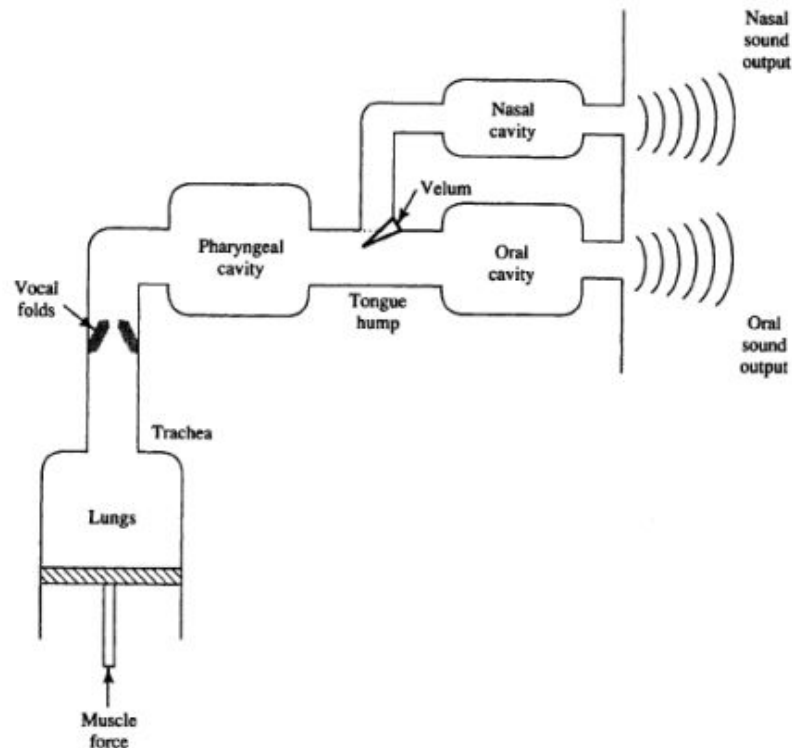
$$S_{xx}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} R_{xx}[k] e^{-j\omega k}$$

Speech Production Mechanism



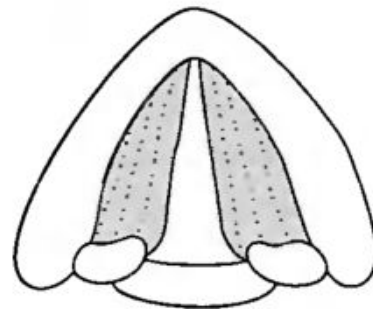
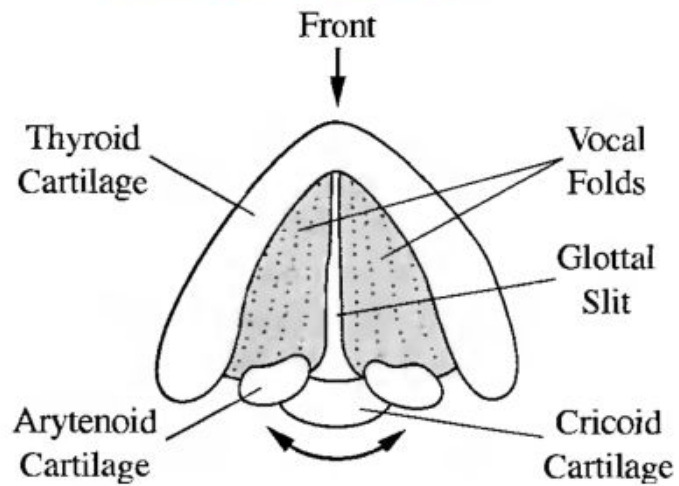
- Speech is the most sophisticated motor activity in the human body
- The motions are lightening fast and totally fluid, yet highly accurate
- Nasal cavity is much larger than oral cavity
- Animation

Block Diagram of Speech Production

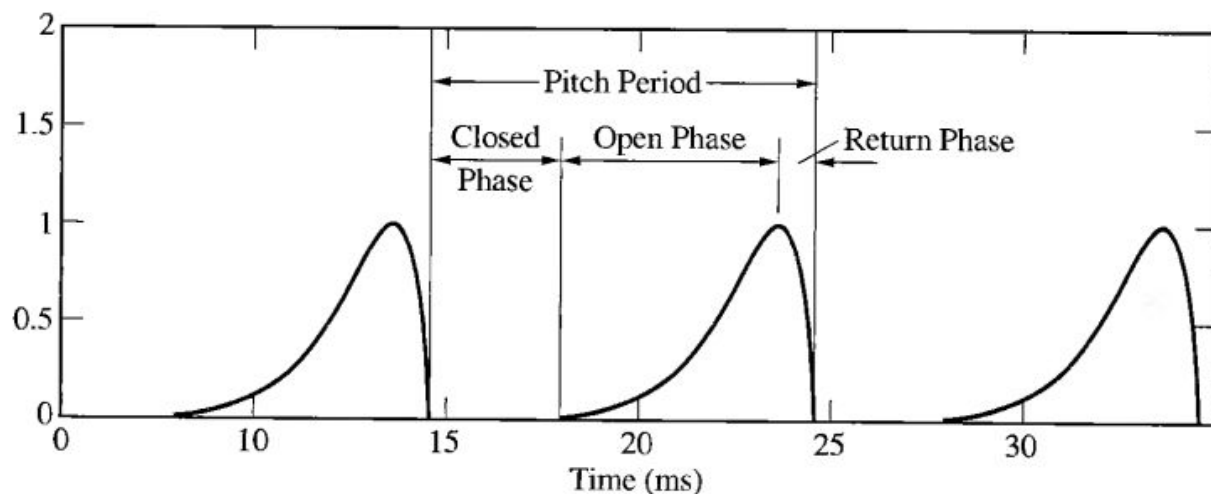


- Lungs act as source of energy
- Vocal folds chop the airflow from lungs into quasi periodic puffs of excitation
- Shape of the vocal tract determines the sound that is produced
- Velum is big enough is decouple nasal cavity, but not oral cavity

Vocal Folds - View & Operation



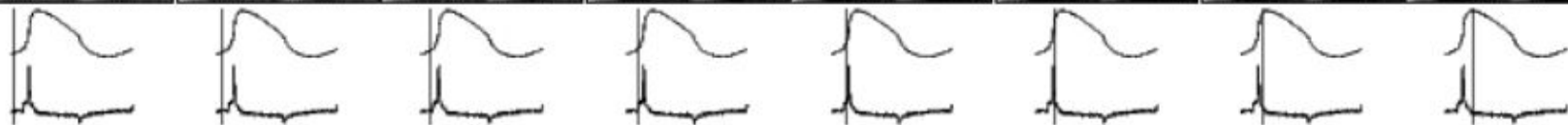
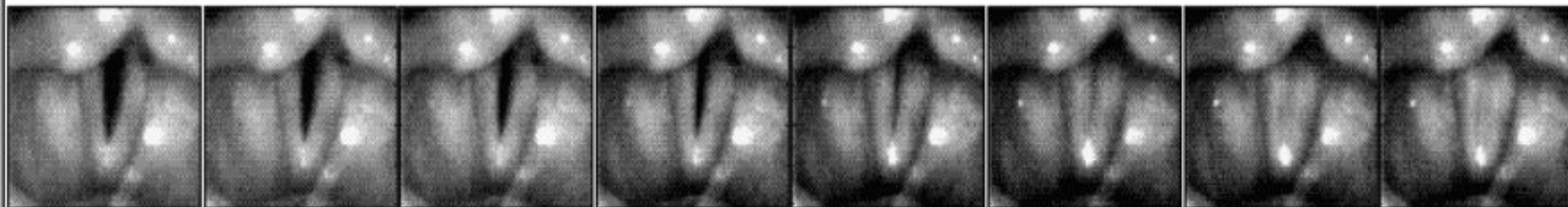
Glottal Flow



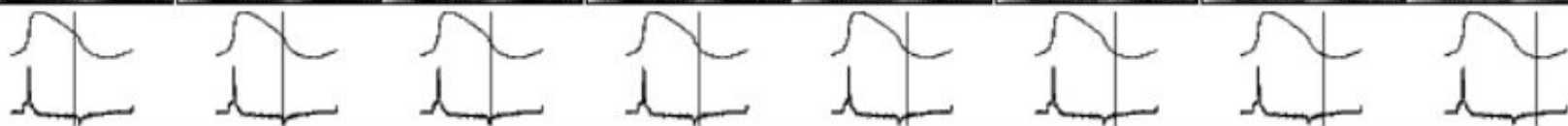
- Vocal folds are muscular tissues in the larynx
- Open when we breathe quietly and closed when we swallow
- Air coming from the lungs causes vocal folds to vibrate between open and closed positions when we talk
- Bernoulli Principle: Increase in the velocity of fluid occurs with simultaneous decrease in the pressure

Laryngeal Cycle

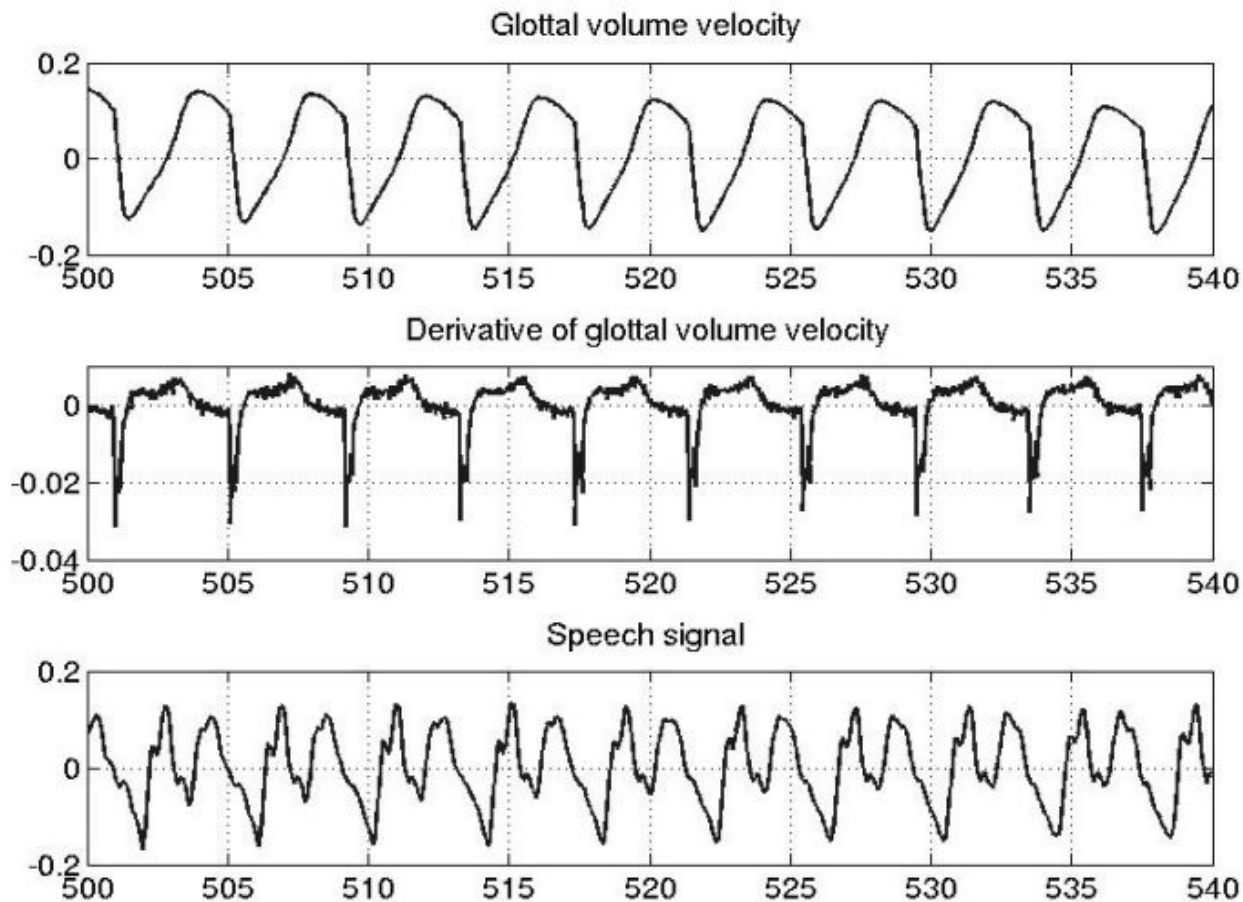
CLOSURE PHASE



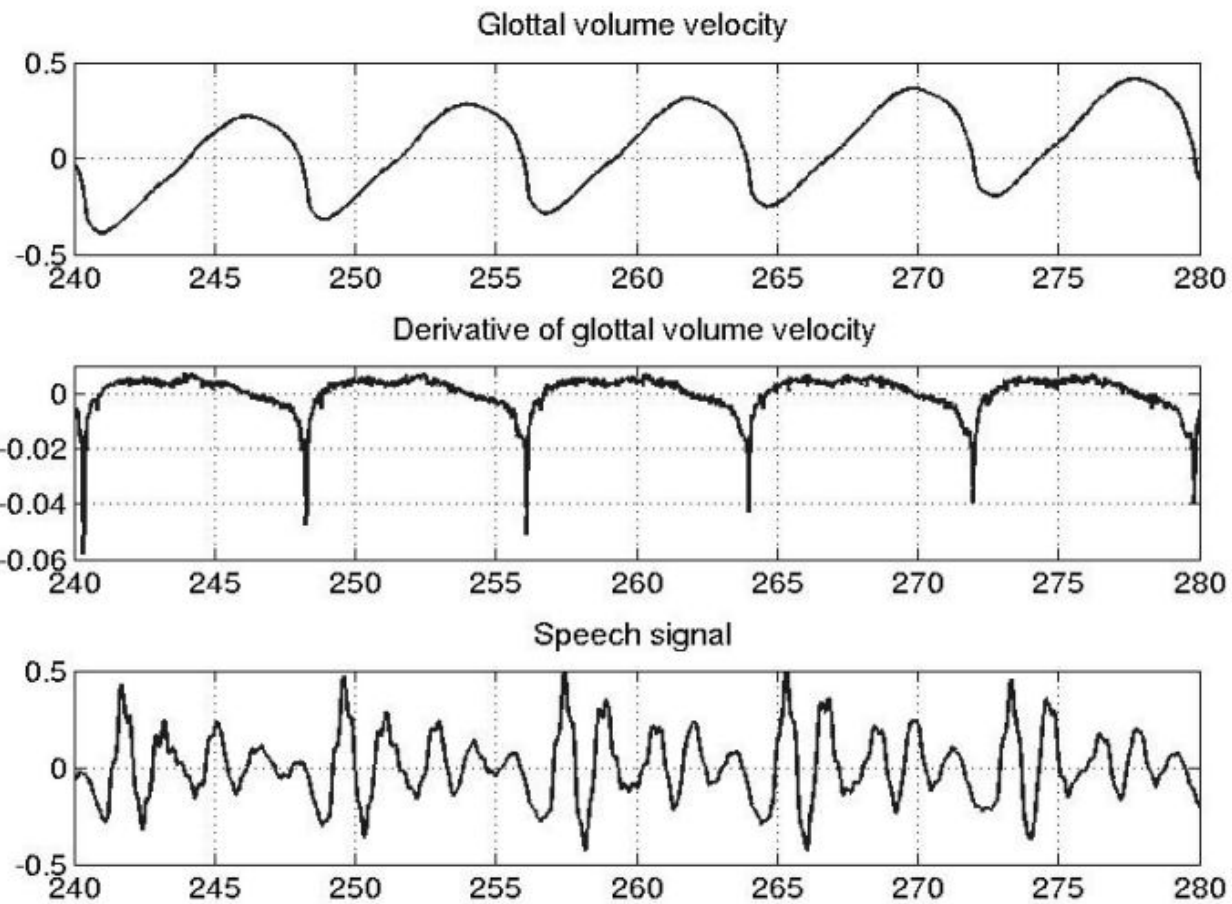
OPENING PHASE



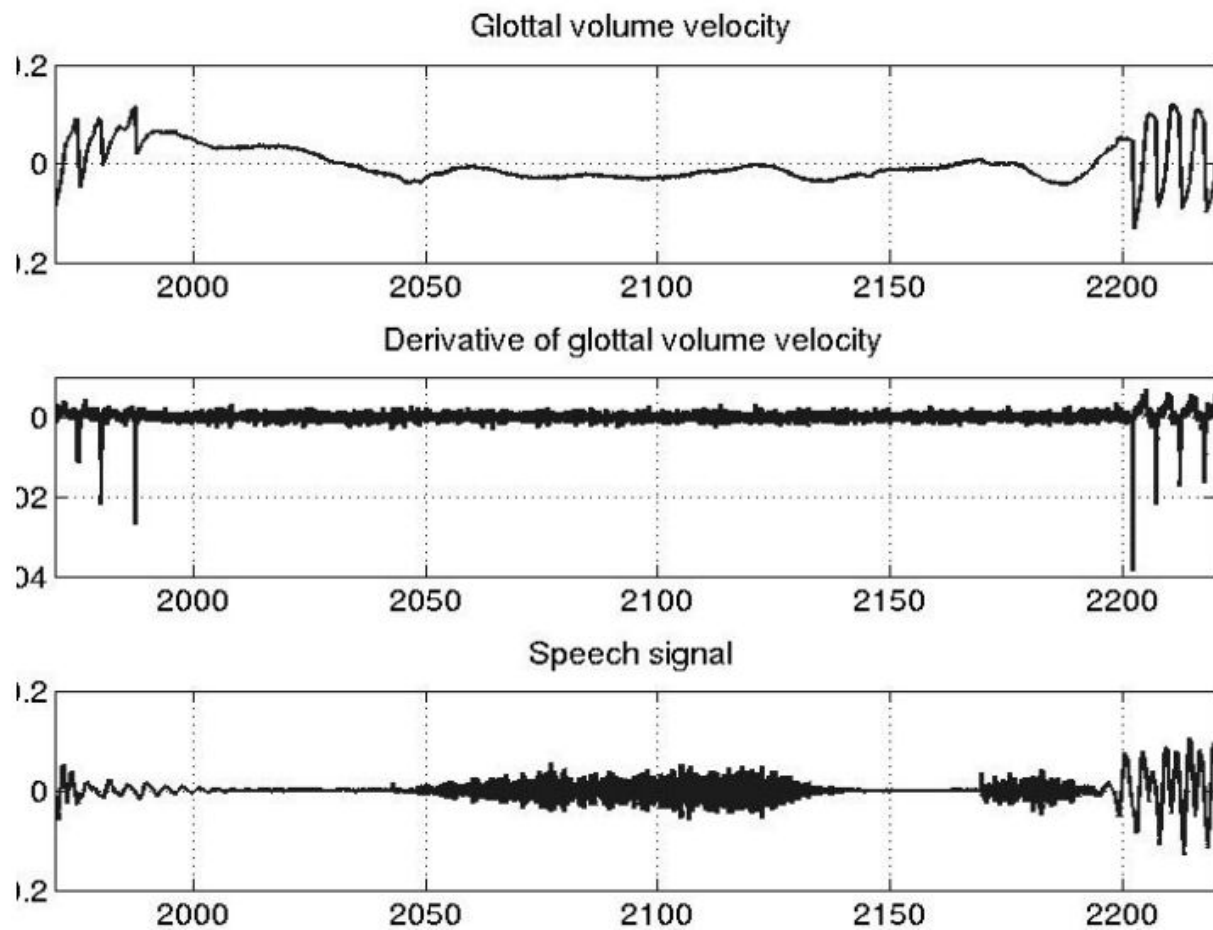
Voice Source - Female Speaker



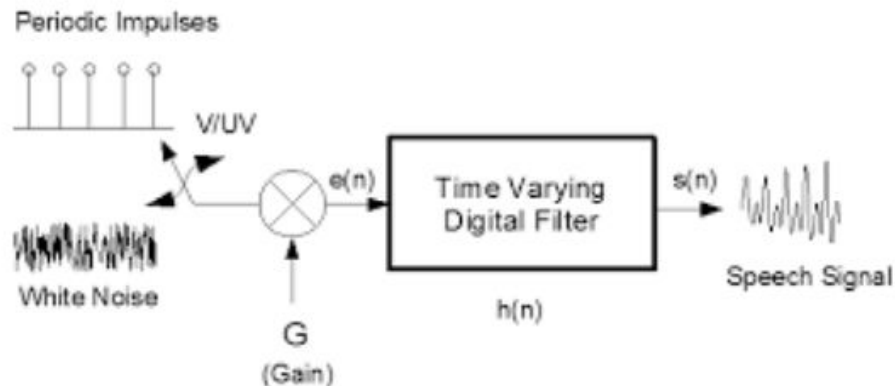
Voice Source - Male Speaker



Voice Source During Frication



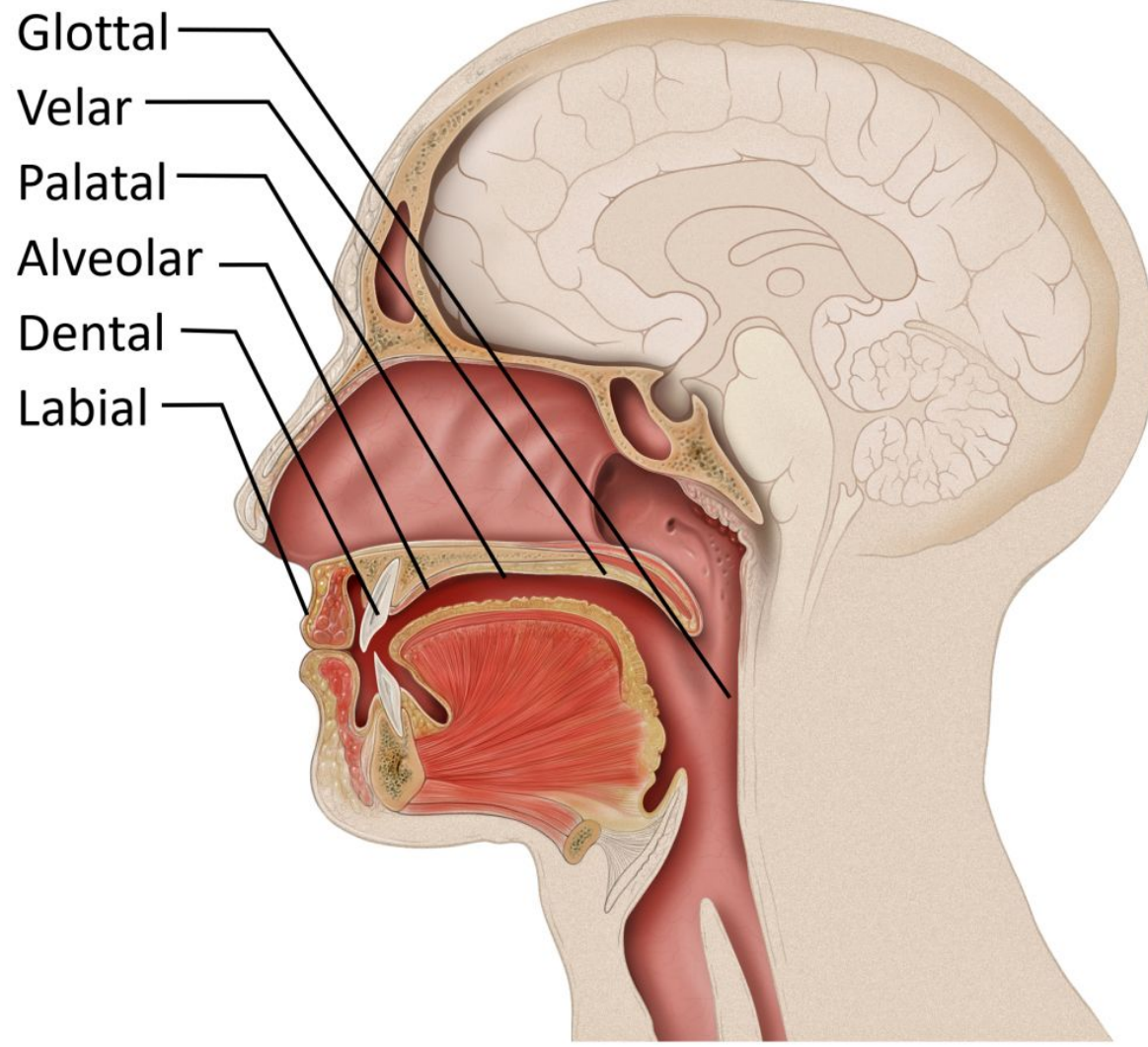
Source-Filter Model of Speech Production



- Speech is an outcome of time-varying vocal tract filter driven by a time-varying excitation
- The state of the vocal cords, the positions, shapes and sizes of the various articulators – all change slowly over time, thereby producing the desired speech sounds
- Need to determine the state of VT from waveform - inverse problem

Speech sounds classification

Manner	Description	Examples
Vowels	Sustained periodic	/i/, /e/, /a/
Plosives (Stops)	Complete closure + burst	/p, b, t, d, k, g/
Fricatives	Narrow constriction → noise	/s, z, ʃ/
Affricates	Stop + fricative	/tʃ, dʒ/
Nasals	Air through nose	/m, n, ŋ/
Liquids	Partial closure	/l, r/
Glides (Semivowels)	Vowel-like	/w, j/ (va,ya)



Place of articulation

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b		t d			ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
Trill	ʙ		r						ʀ		
Tap or Flap		ⱱ	ɾ			ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative			ɬ ɮ								
Approximant		ʋ	ɹ			ɻ	j	ɰ			
Lateral approximant			l			ɭ	ʎ	ʟ			

Initial vowels											
అ	ఆ	ఇ	ఐ	ఉ	ఊ	ఋ	ౠ	ఎ	ఏ	ఐ	ఒ
a	ā	i	ī	u	ū	r̥	e	ē	ai	o	ō
[a]	[a:/æ:]	[i]	[i:]	[u]	[u:]	[r̥]	[e]	[e:]	[aj]	[o]	[o:]

క ఖ గ ఘ ఙ

ka kha ga gha ṅa
[ka] [kʰa] [ga] [gʱa] [ŋa]

Velar stops and nasal

చ ఛ జ ఝ ఞ

ca cha ja jha ṇa
[tʃa] [tʃʰa] [dʒa] [dʒʱa] [ɲa]

Palatal affricates and nasal

ట ఠ డ ఢ ణ

ṭa ṭha ḍa ḍha ṇa
[ṭa] [ṭʰa] [ḍa] [ḍʱa] [ɳa]

Retroflex stops and nasal

త థ ద ధ న

ta tha da dha na
[ta] [tʰa] [da] [dʱa] [na]

Dental stops and nasal

ప ఫ బ భ మ

pa pha ba bha ma
[pa] [pʰa/fa] [ba] [bʱa] [ma]

Labial stops and nasal

య ర ల వ ళ

ya ra la va ḷa
[ja] [ra] [la] [wa] [ḷa]

Semivowels (liquids and glides)

శ ష స హ

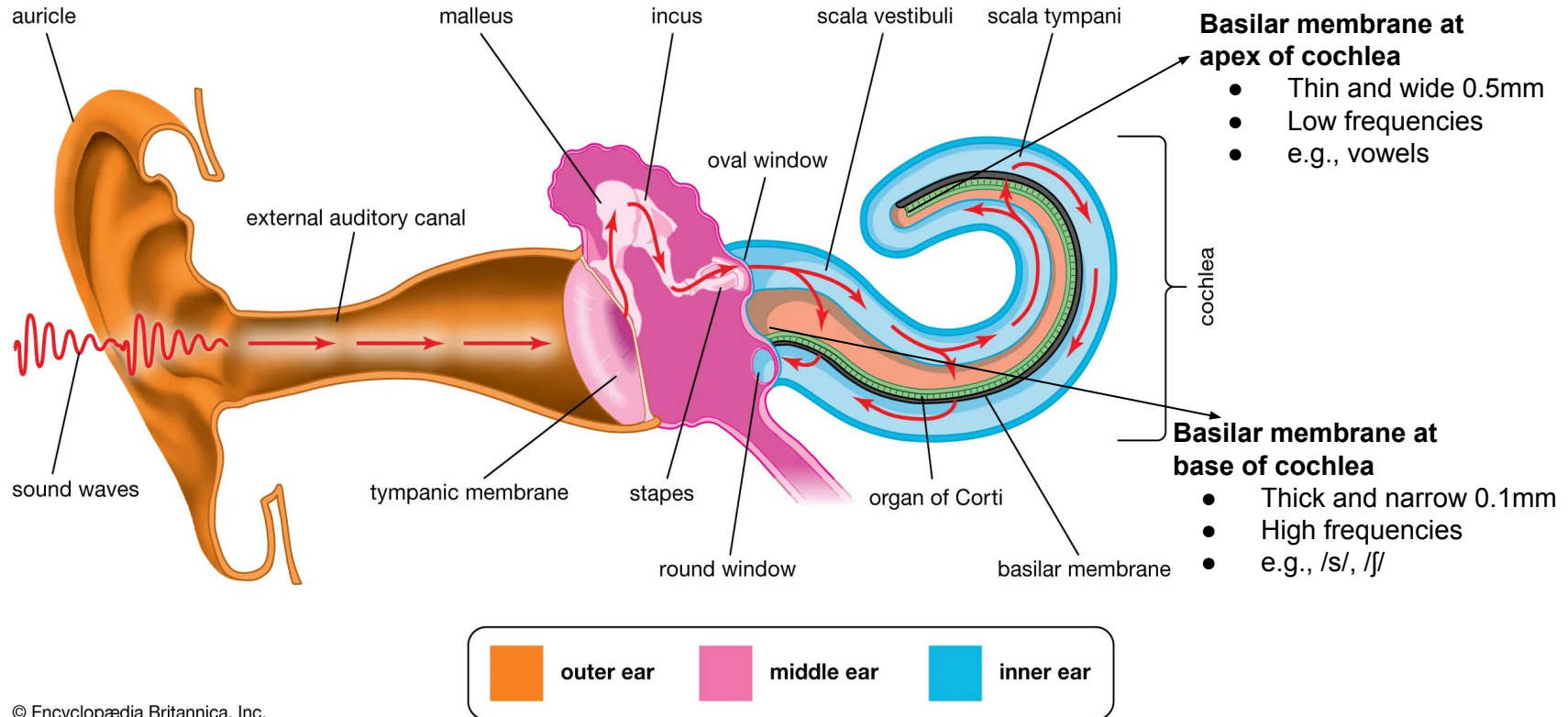
śa ṣa sa ha
[ʃa] [ʂa] [sa] [ha]

Fricatives

Audio perception by ear

engineered as **mechanical**—**hydraulic**—neural chain.

BM length is typically 32–35 mm



Basilar membrane: a tuned band-pass filter

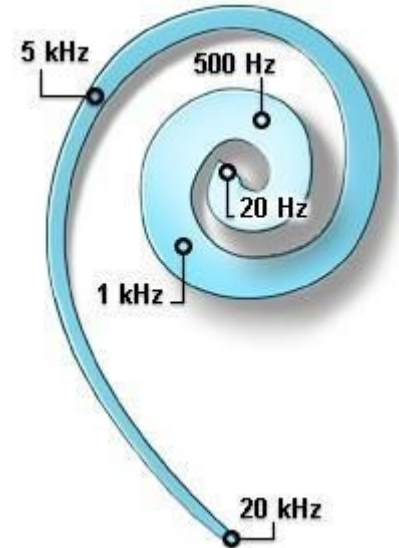
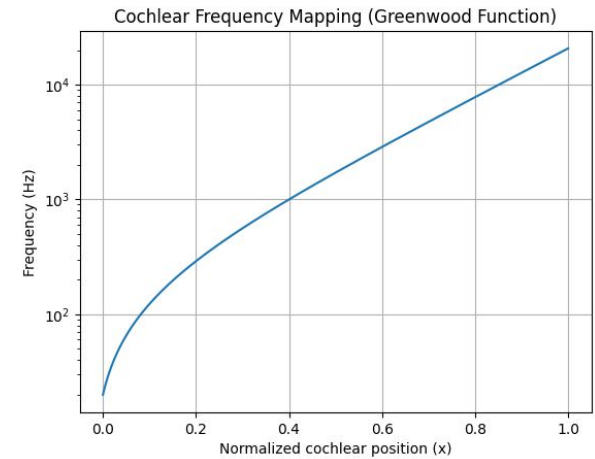
Frequency along BM: $f(x) = 165.4 (10^{2.1x} - 0.88) = 20 \text{ Hz}; x=0 \text{ (apex)}$
 $= 20,000; x=1 \text{ (base)}$

Differential of $f(x)$: $f'(x) = K \cdot 10^{(2.1 \cdot x)}$

Large near base $\rightarrow$ poor spatial resolution : at high frequencies

Small near apex $\rightarrow$ fine resolution: at low frequencies

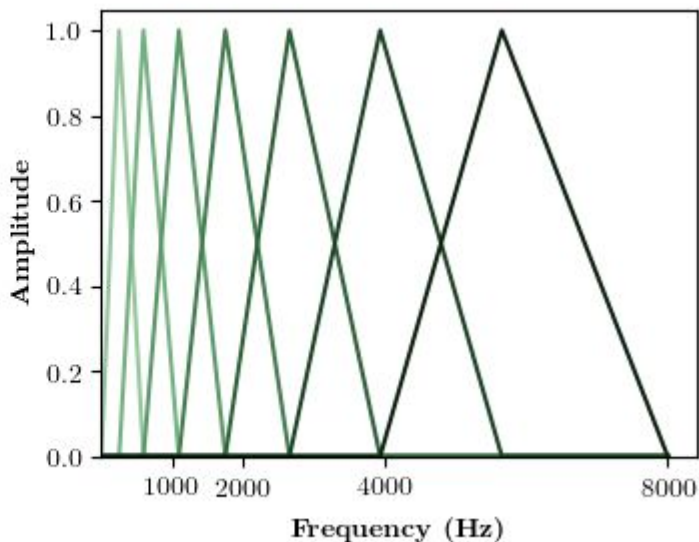
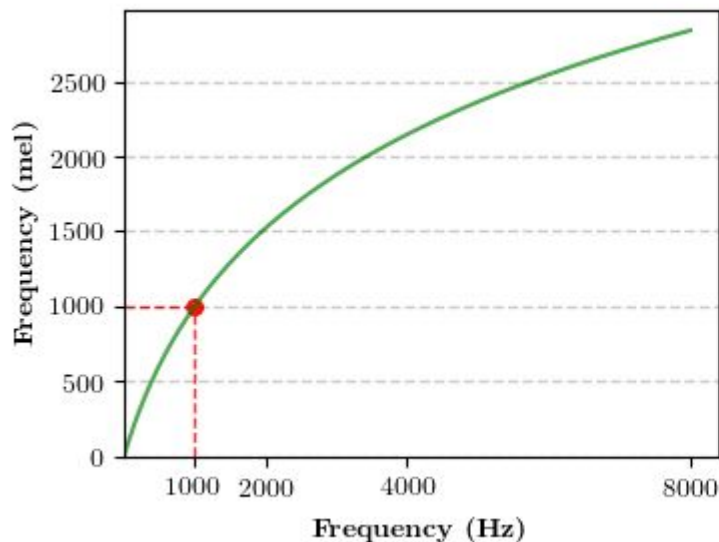
Cochlear Position	Approx. Frequency Range	Speech Relevance
Base	8–20 kHz	Fricatives (/s/, /f/)
Middle	500 Hz – 4 kHz	Most speech intelligibility
Apex	20–500 Hz	Pitch, vowels



Ear does not hear frequencies linearly

- Linear at low frequencies
- Logarithmic at high frequencies
- Instead of uniform Frequency bins, map linear frequencies $\rightarrow$ cochlear positions: Given as **mel-scale**

$$\text{mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$



Sound units in an Indian Language

Characters representing vowel sounds

Short vowels	/a/(अ)	/i/(इ)	/u/(उ)	/e/(ए)	/o/(ओ)
Long vowels	/a:/(आ)	/i:/(ई)	/u:/(ऊ)	/e:/(ऎ)	/o:/(औ)
Diphthongs	/ai/(ऐ)		/au/(औ)		

Characters representing Consonant-Vowel (CV) combinations with vowel /a/(अ)

Place of articulation	Manner of articulation				Nasals	Semivowels	Fricatives
	Unvoiced		Voiced				
	Unaspirated	Aspirated	Unaspirated	Aspirated			
Velar	/ka/(क)	/kha/(ख)	/ga/(ग)	/gha/(घ)	/kna/(ङ)	----	/ha/(ह)
Palatal	/cha/(च)	/chha/(छ)	/ja/(ज)	/jha/(झ)	/chna/(ञ)	/ya/(य)	/sha/(श)
Retroflex	/ṭa/(ट)	/ṭha/(ठ)	/ḍa/(ड)	/ḍha/(ढ)	/ṭna/(ण)	/ra/(र)	/shha/(ष)
Dental	/ta/(त)	/tha/(थ)	/da/(द)	/dha/(ध)	/na/(न)	/la/(ल)	/sa/(स)
Bilabial	/pa/(प)	/pha/(फ)	/ba/(ब)	/bha/(भ)	/ma/(म)	/va/(व)	----

Short time analysis of Speech

- A phoneme (like /a/) doesn't have a single, static acoustic property.
- Its production involves continuous movement of the articulators (tongue, lips, jaw).
- Therefore, the acoustic signal changes *within* the phoneme.

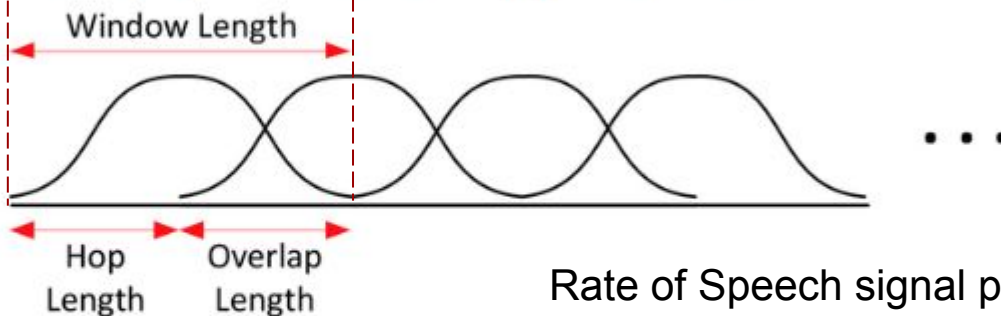
Short time analysis

- The continuous speech waveform is divided into short, overlapping frames (typically 20-25 ms long, with a 10 ms shift between frames).
- Each frame is assumed to be quasi-stationary (statistically stable within that short window).

Signal



Windows



**Segmentation
/ Windowing**



Rate of Speech signal processing

$$\text{Hop Length} = \text{Window Length} \times (1 - \text{Overlap}\%)$$

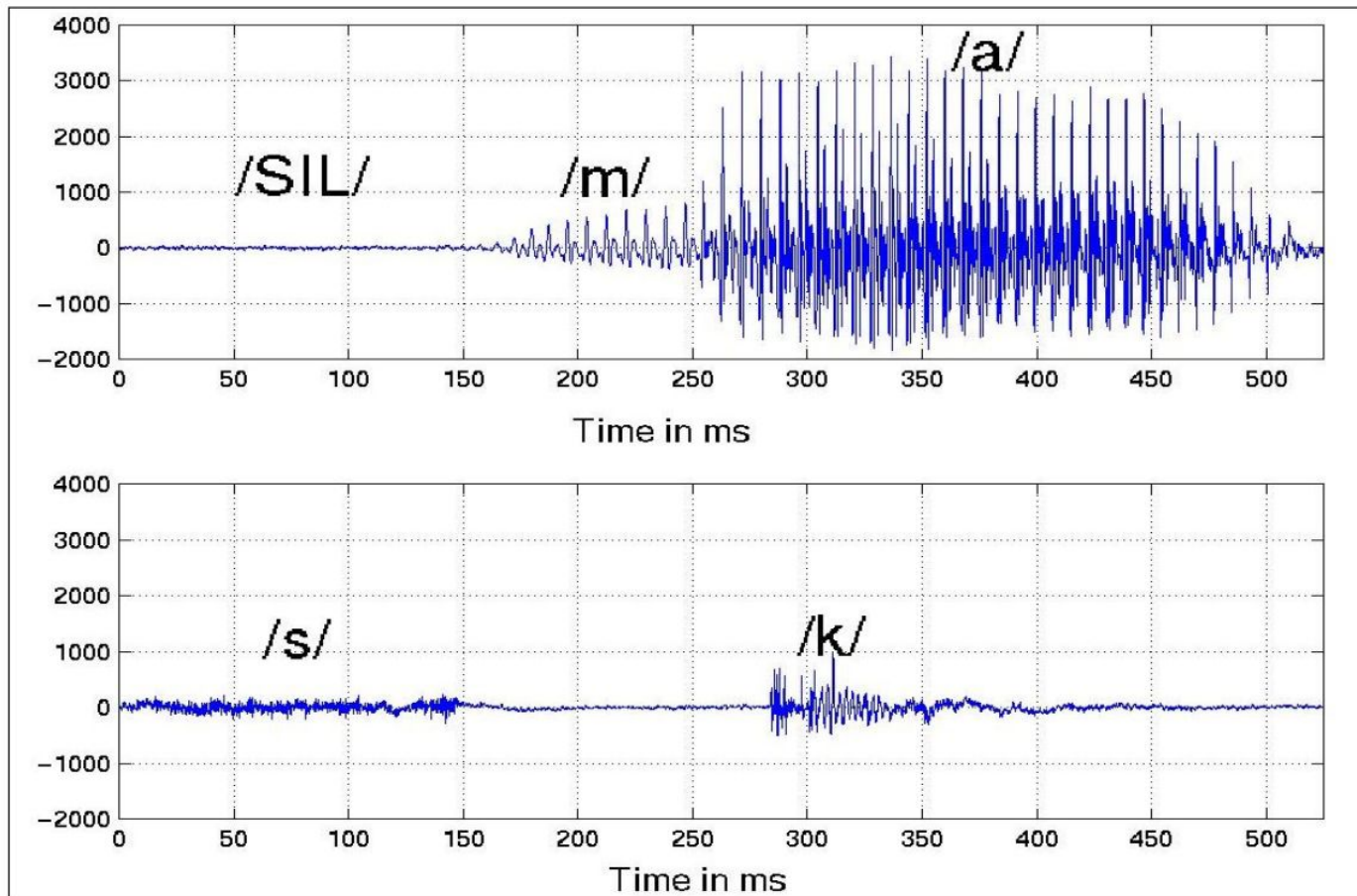
$$\text{Overlap}\% = (1 - \text{Hop Length} / \text{Window Length}) \times 100$$

$$\begin{aligned} \text{Frame Rate} &= \text{Sample Rate} / \text{Hop Length (in samples)} \\ &= 1 / \text{Hop Length (in sec)} \end{aligned}$$

Compute frame rate of a speech signal with $f_s=16\text{KHz}$, 20ms window and 25% frame overlap.

Short time speech analysis

m/a/s/k
n/v/f/stop



Speech Signal Processing

- Constraints of production at various levels in the signal
- Gross level features: Periodic, UV, burst, sil
- Source-System model
- Subsegment level (within a frame $< 40\text{ms}$): frame level features, MFCC
- Segment level (50-200ms): V/UV, Pitch, formants
- Suprasegment level ($>200\text{ms}$ /utterance level): F0 contours, duration, coarticulation
- Time-freq. representation: Temporal, spectral, spectrograms

Energy of a signal

<i>Property</i>	<i>Continuous</i>	<i>Discrete</i>
Energy	$E_x = \int_{-\infty}^{\infty} x(t) ^2 dt < \infty$	$E_x = \sum_{n=-\infty}^{\infty} x[n] ^2$
Power	$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) ^2 dt < \infty$	$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{n=-\infty}^{\infty} x[n] ^2 < \infty$

Short time energy:

Energy of the m-th frame

$$E_m = \sum_{n=1}^N |x[n]w[n - mh]|^2$$

End points detection

Word recognition needs to detect word boundaries in speech vs recognizing silence.

- Weak fricatives: /f, th, h/
- Weak plosive bursts: /p, t, k/
- Final nasals
- Voiced fricatives becoming devoiced

Solution

Two measurements: Energy and Zero crossing rate

Energy

- **High energy:** vowels, voiced sounds
- **Low energy:** silence, unvoiced sounds

Zero crossing Rate:How frequently the signal changes sign (positive ↔ negative)

$$ZCR_m = \frac{1}{2N} \sum_{n=1}^N |\text{sgn}(x_m[n]) - \text{sgn}(x_m[n-1])|$$

- **High ZCR** → unvoiced sounds (/s/, /f/), noise
- **Low ZCR** → voiced sounds (vowels)

Condition	Decision
$(E(m) < E_{th})$	Silence / Non-speech
$(E(m) > E_{th}) \text{ AND } (ZCR(m) < Z_{th})$	Voiced speech
$(E(m) > E_{th}) \text{ AND } (ZCR(m) > Z_{th})$	Unvoiced speech

From silence regions estimate mean and variance of energy and zcr

- Energy threshold:

$$E_{th} = \mu_E^{silence} + \alpha \sigma_E$$

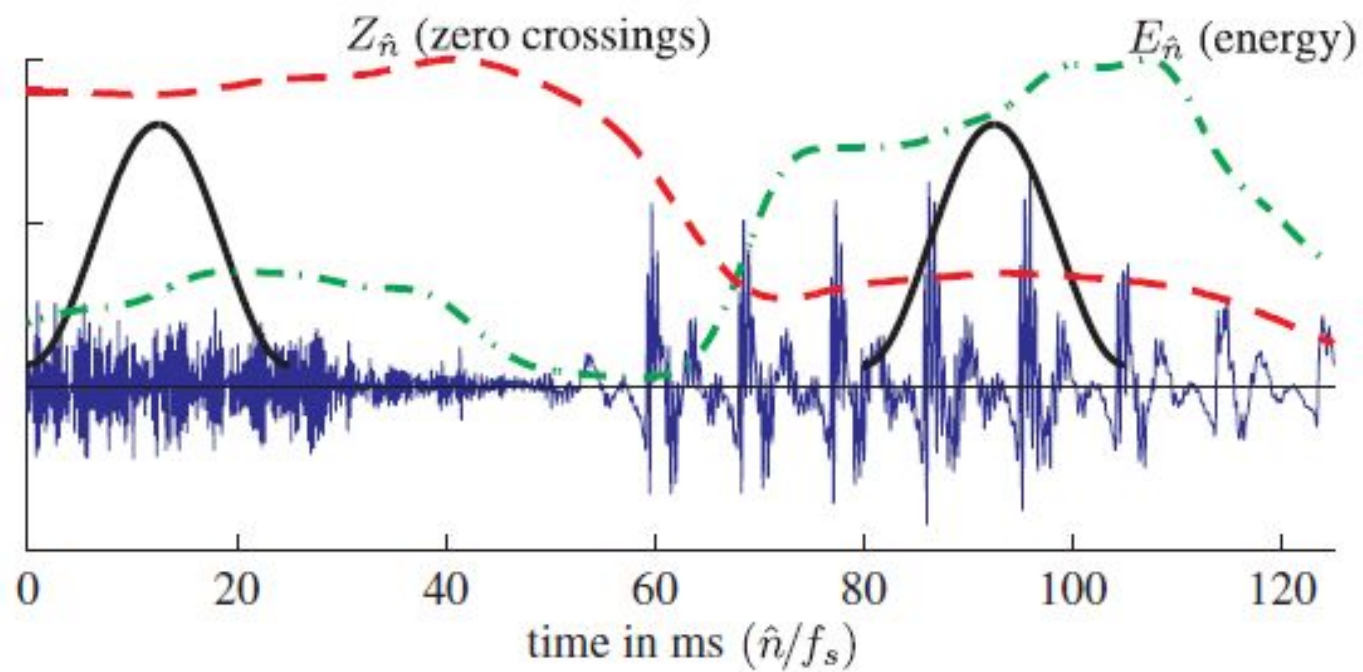
- ZCR threshold:

$$Z_{th} = \mu_{ZCR}^{silence} + \beta \sigma_{ZCR}$$

Decision for word end points

Word endpoints are detected by first locating high-energy regions to identify speech segments and then refining the start and end boundaries using zero-crossing rate.

Region	Energy	ZCR
Silence	Low	Low
Word onset	Low → High	High
Voiced part	High	Low
Word offset	High → Low	High



RABINER and SAMBUR algorithm for word end points

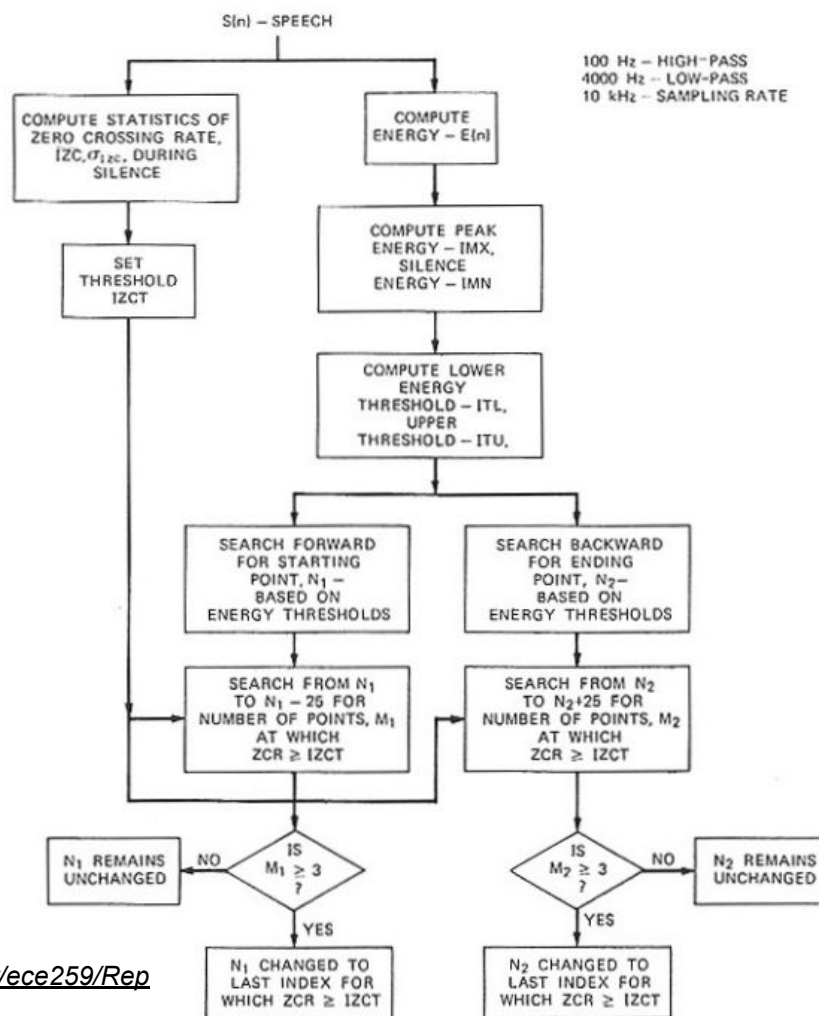


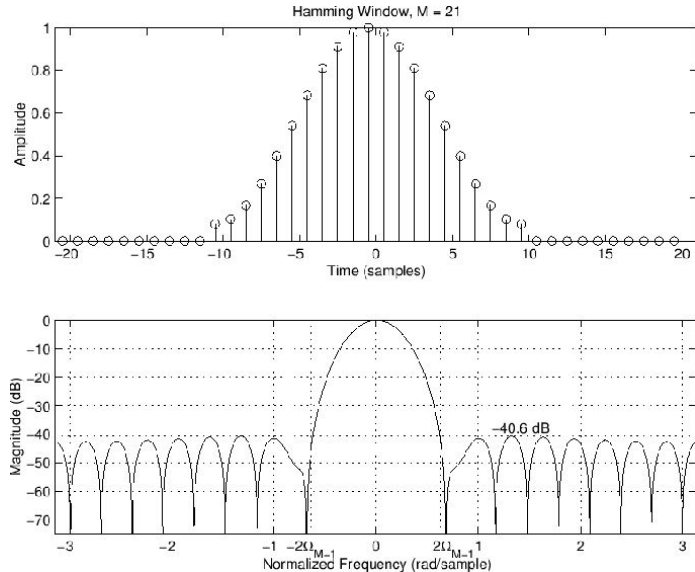
Fig. 9—Flowchart for the endpoint algorithm.

Short time Fourier transform

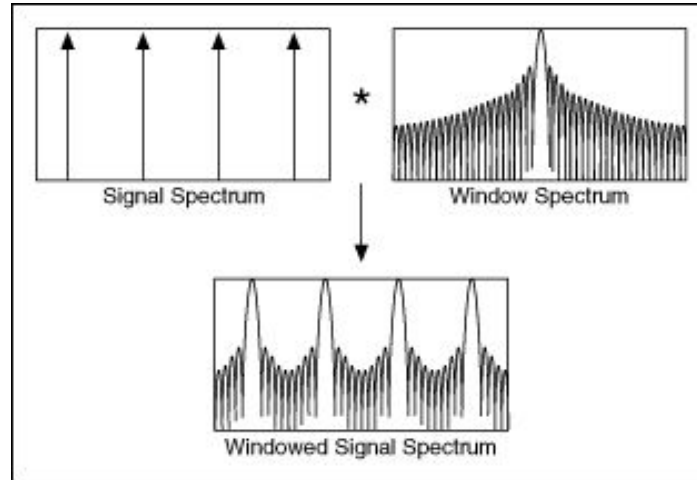
F.T of mth frame

$$X_m(\omega) = \sum_{n=-\infty}^{\infty} x(n)w(n - mR)e^{-j\omega n}$$

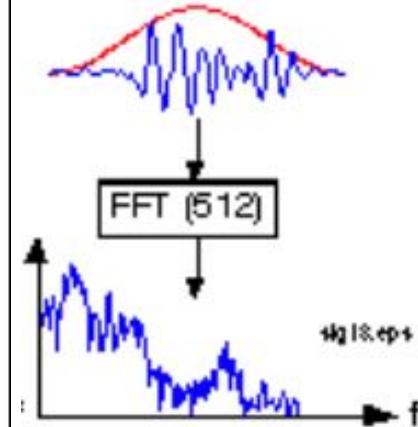
Window and its spectrum

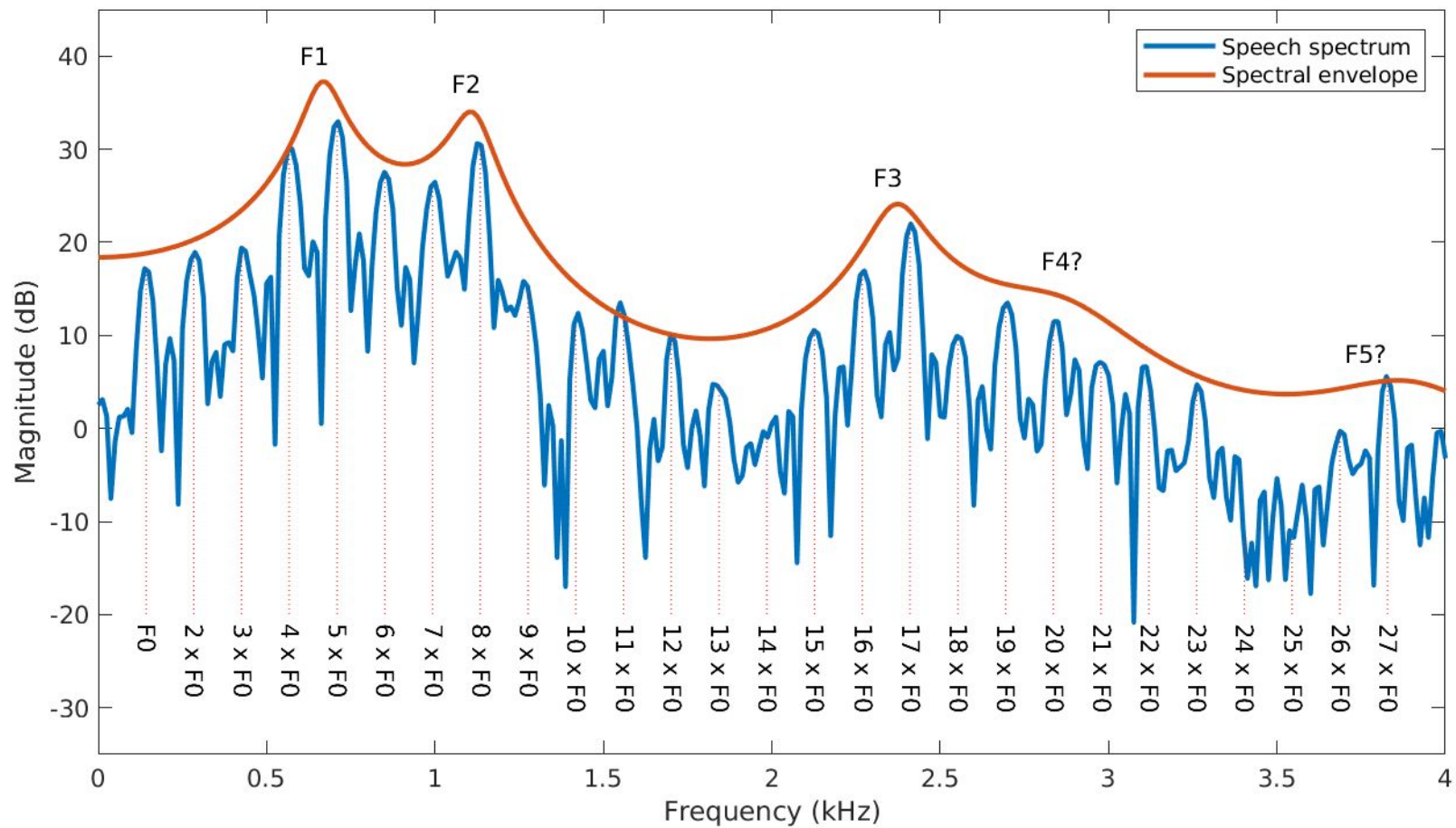


Effect of Windowing in frequency



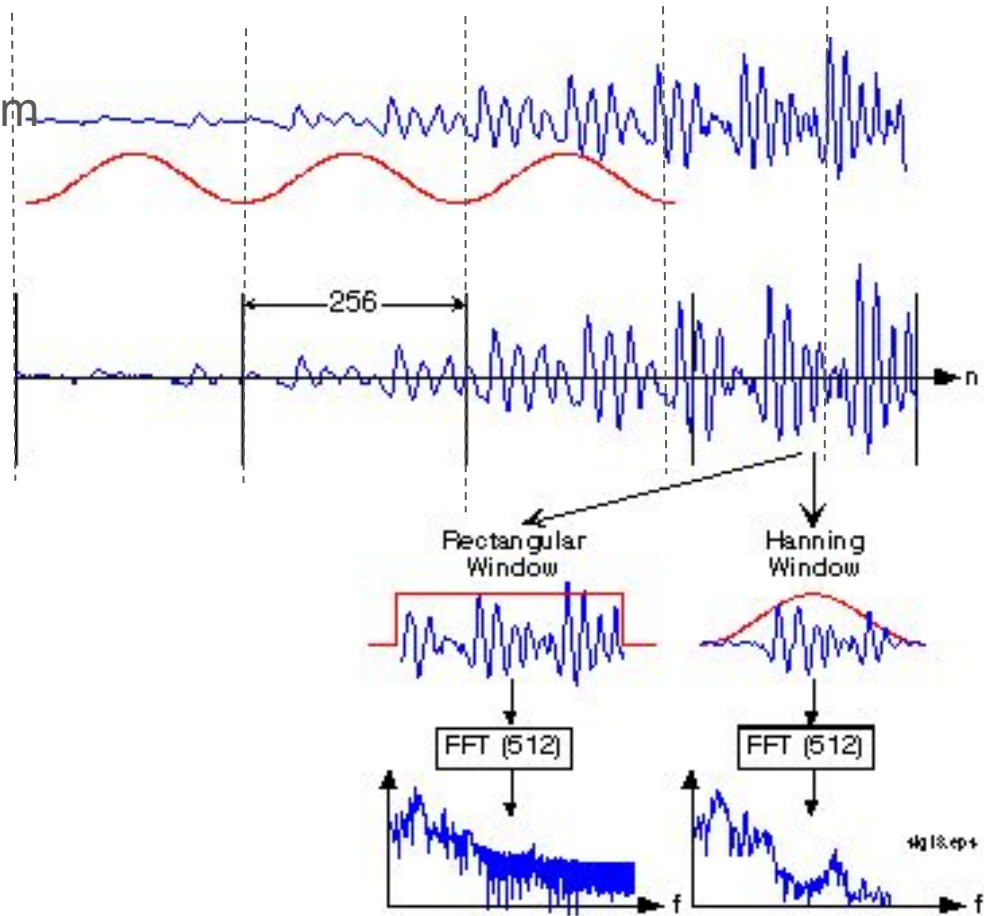
Windowed speech and its spectrum

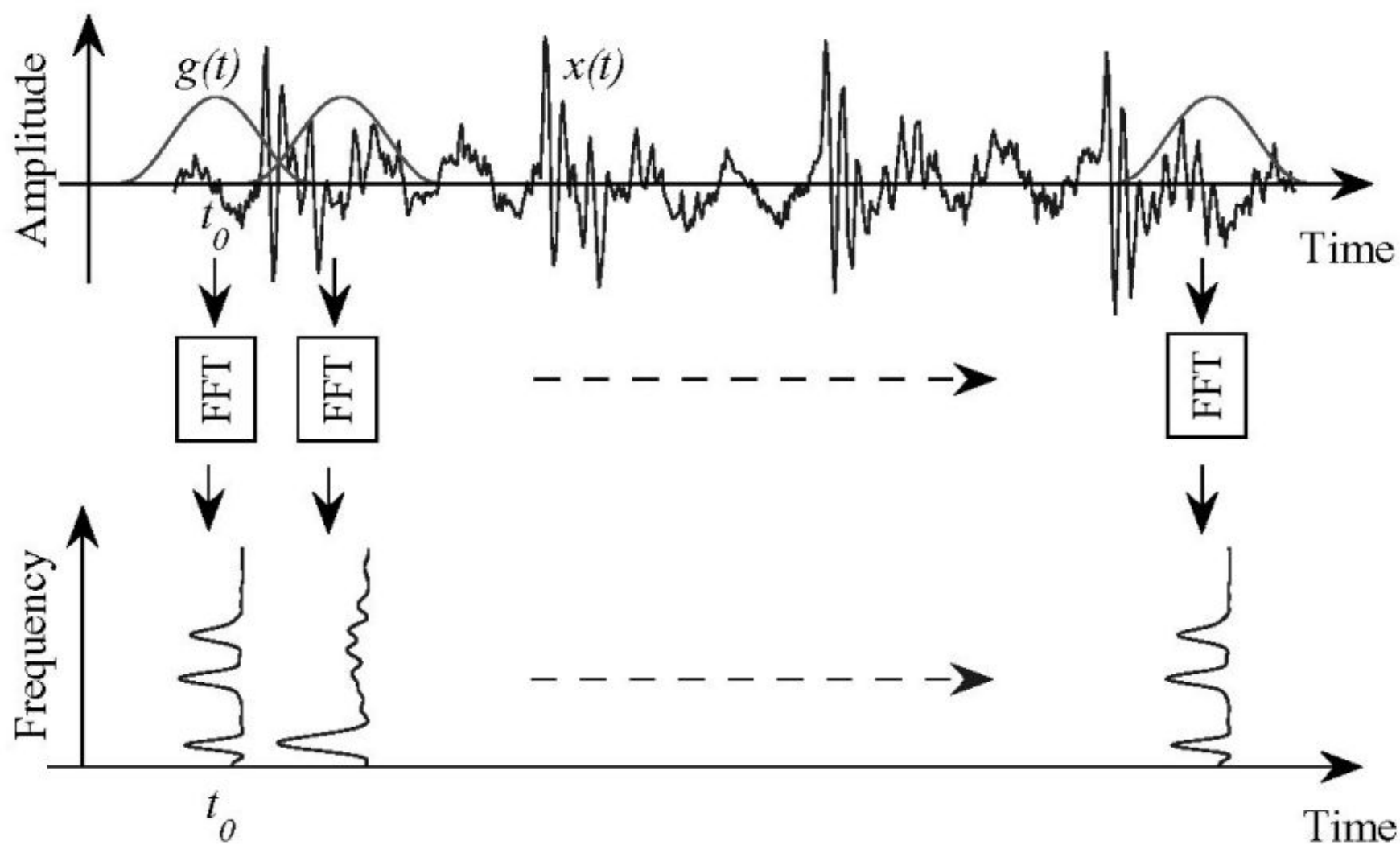


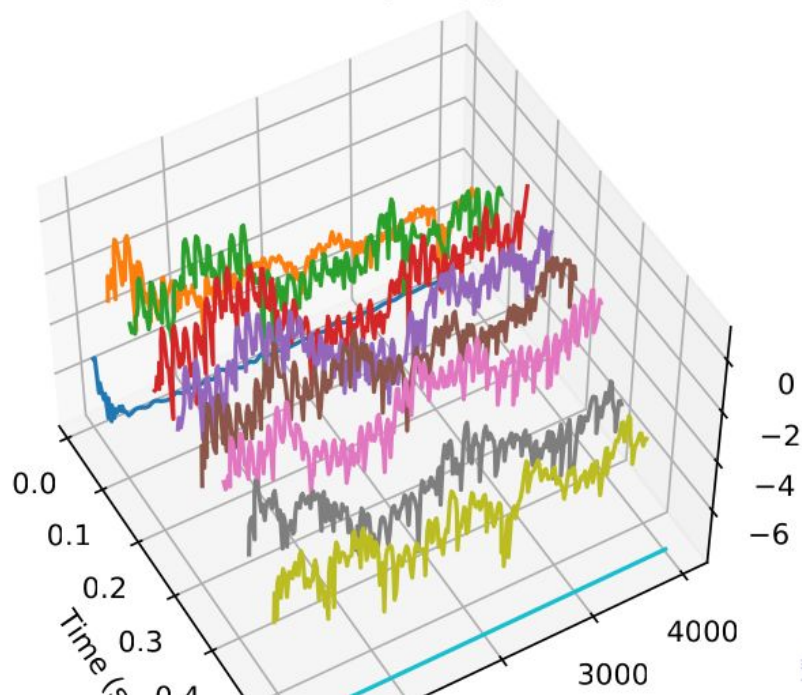
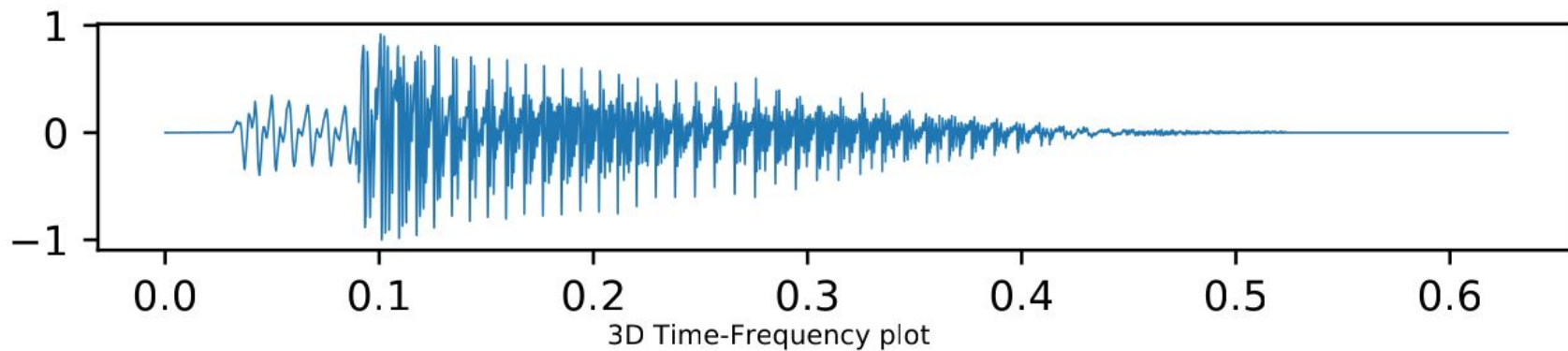


Speech waveform

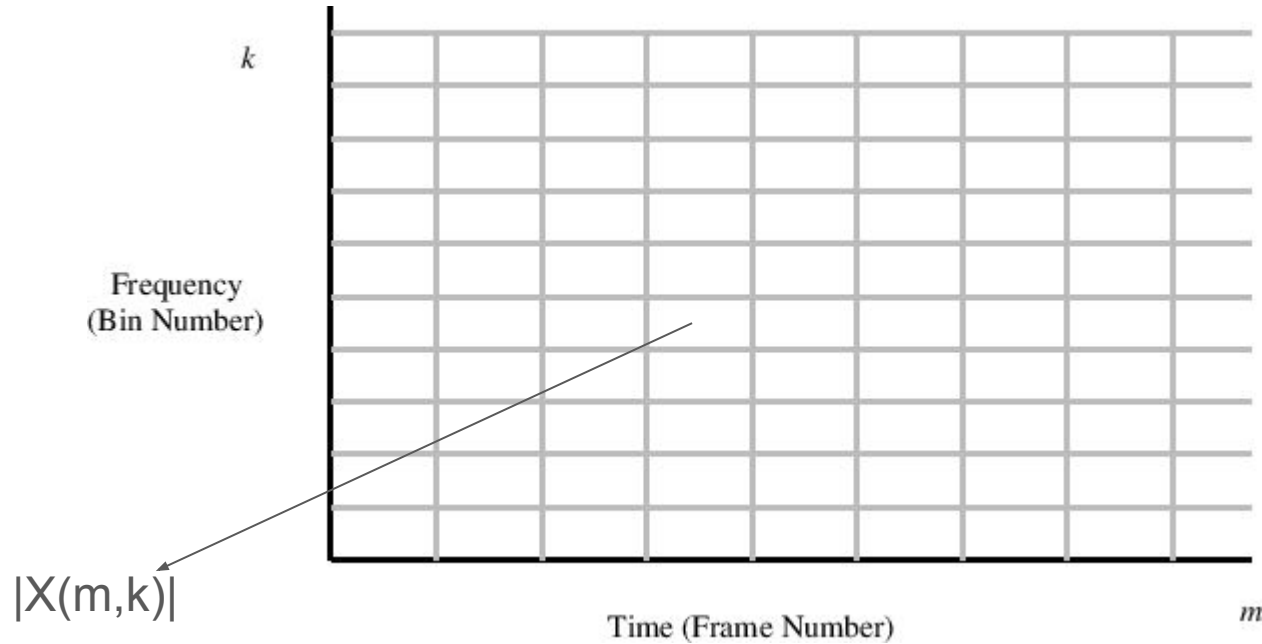
windowing





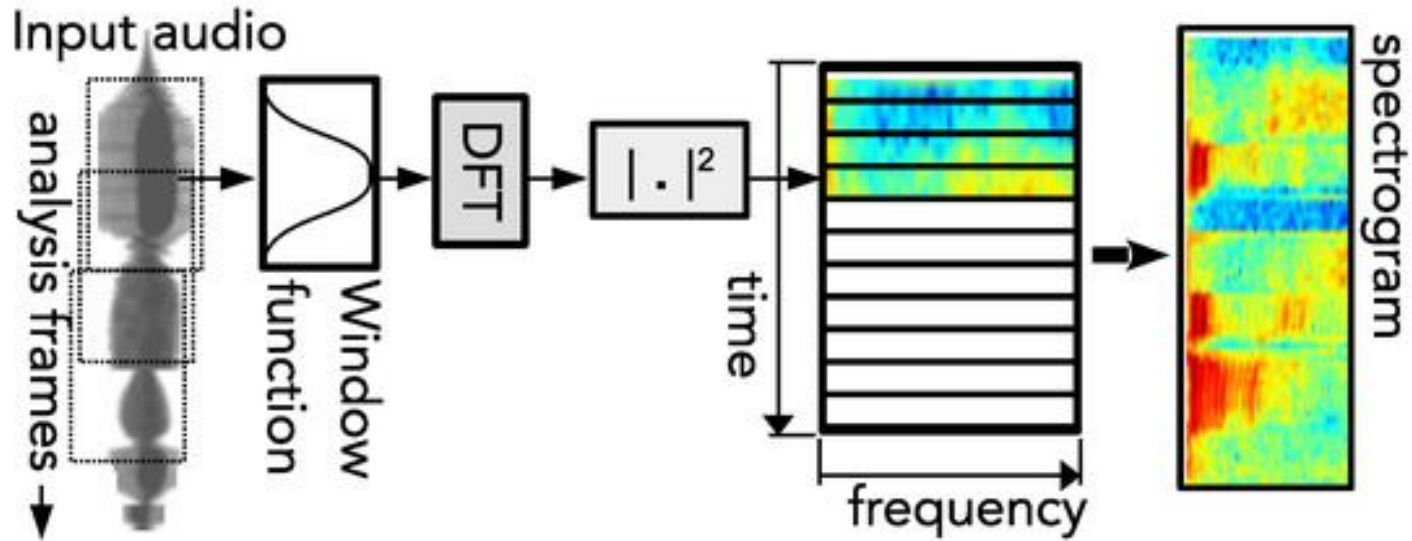


Spectrogram (magnitude)

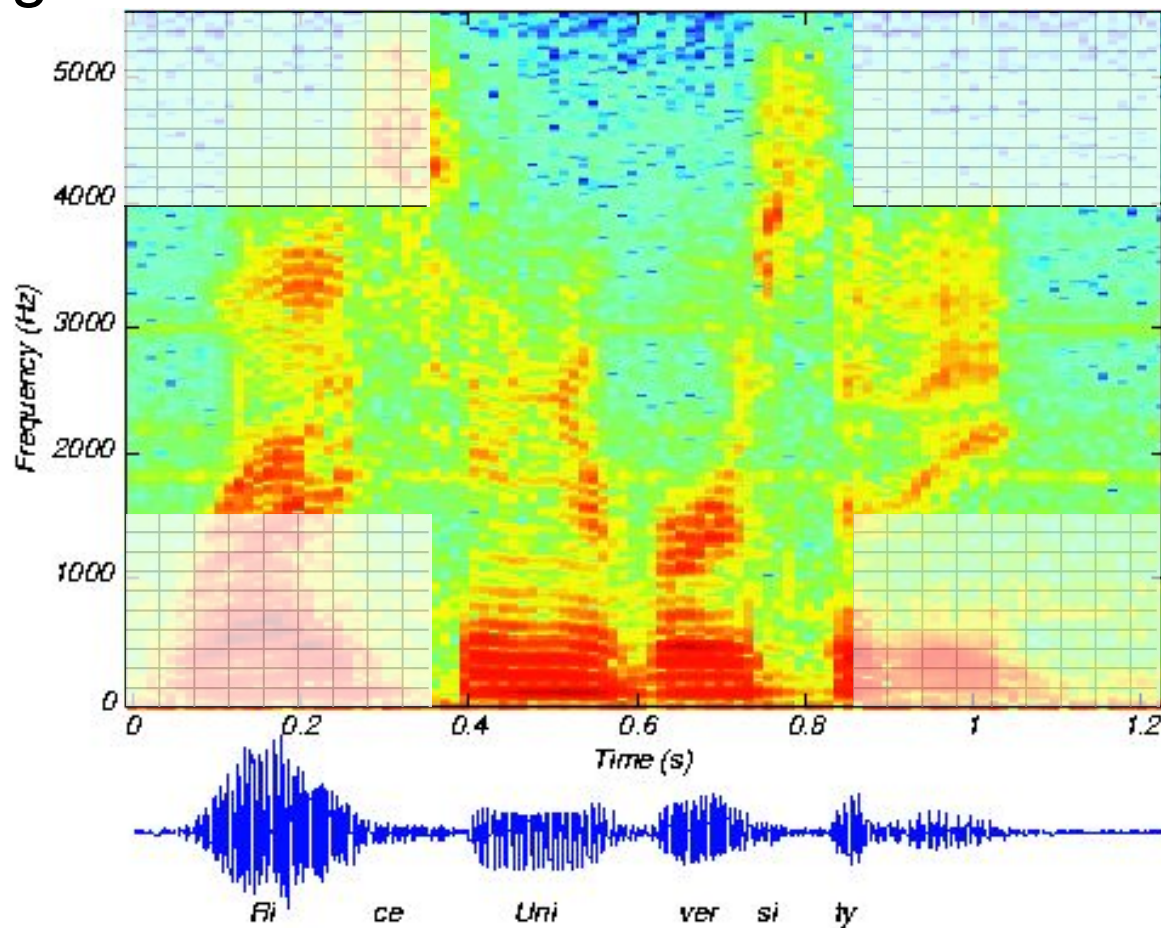


Magnitude of $X_m(w)$ at $w=2*\pi*k/N$

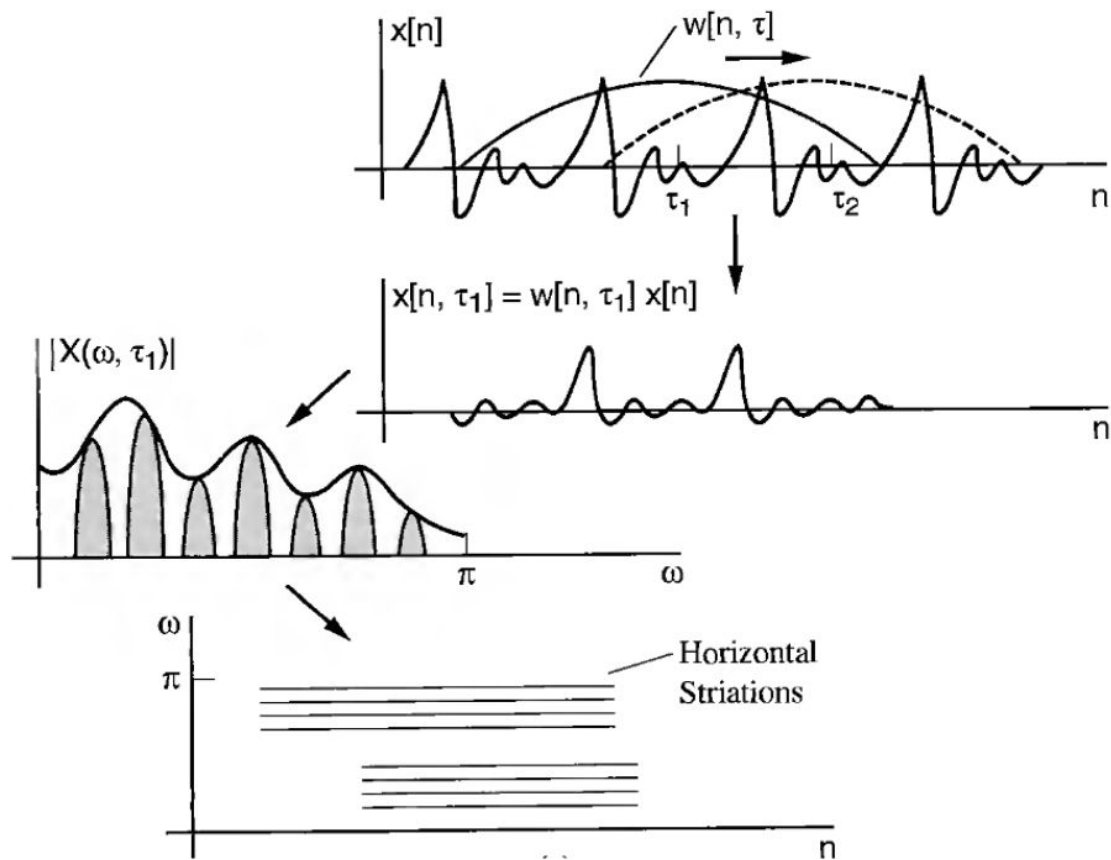
Spectrogram



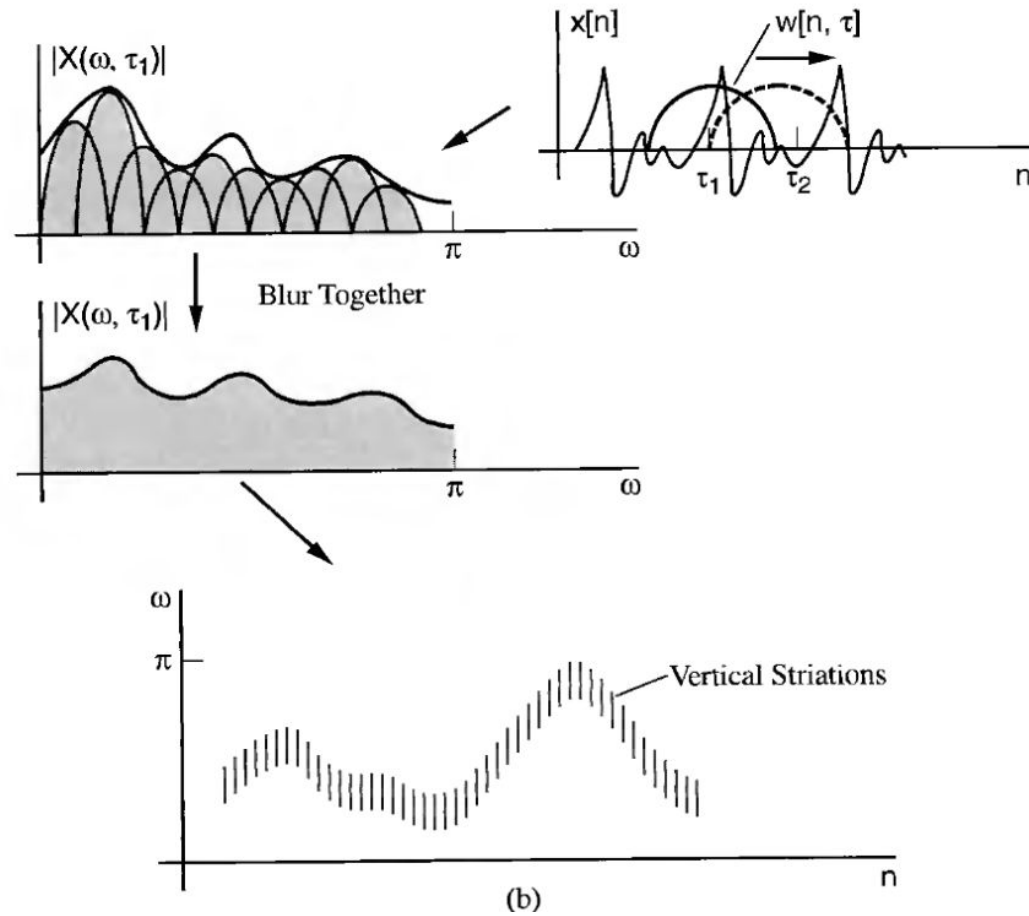
Spectrogram



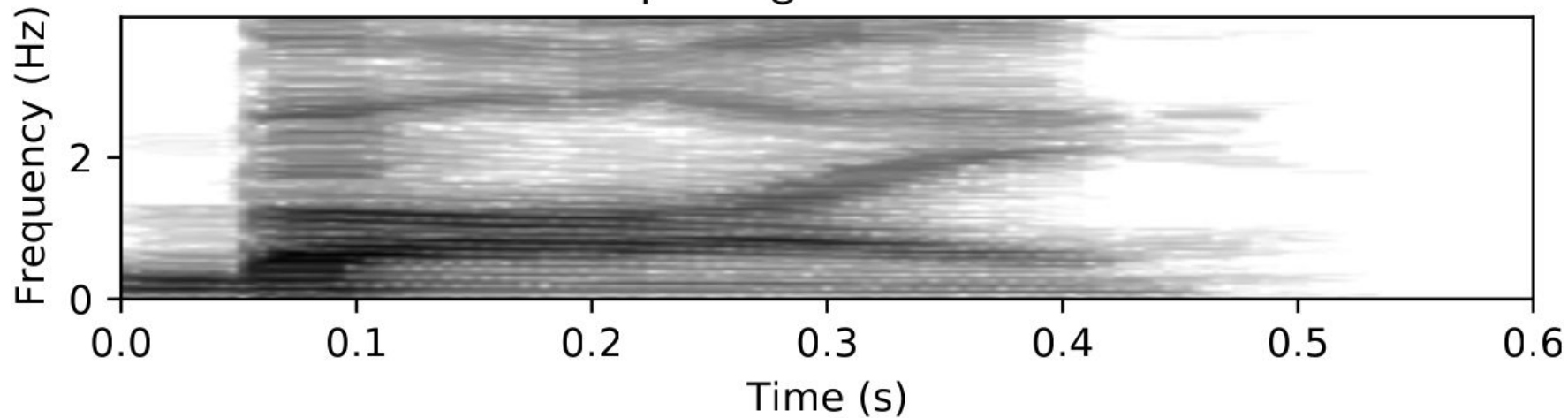
Narrow band spectrogram



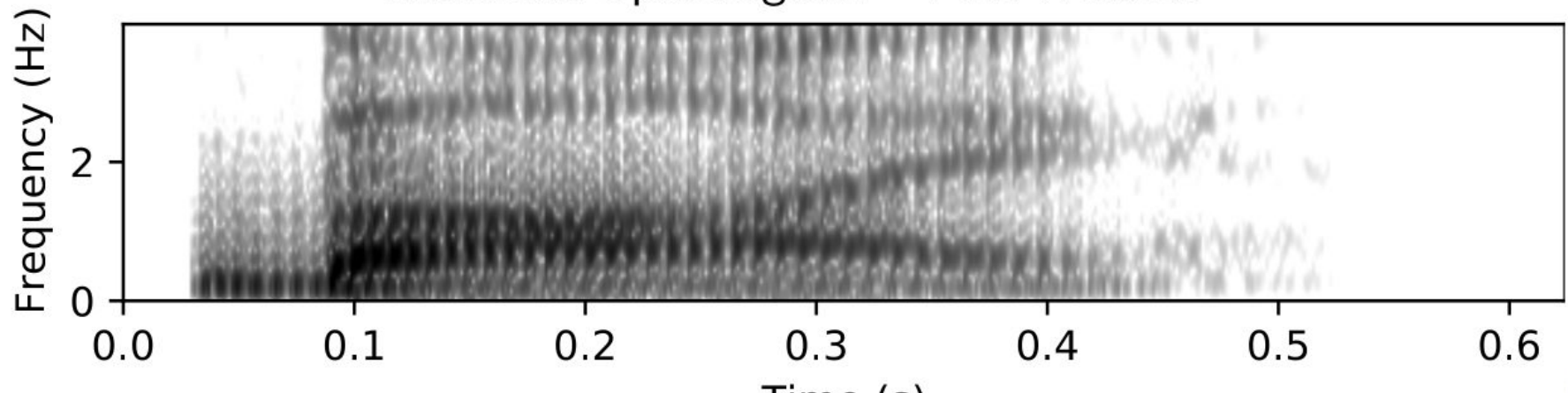
Wide band spectrogram

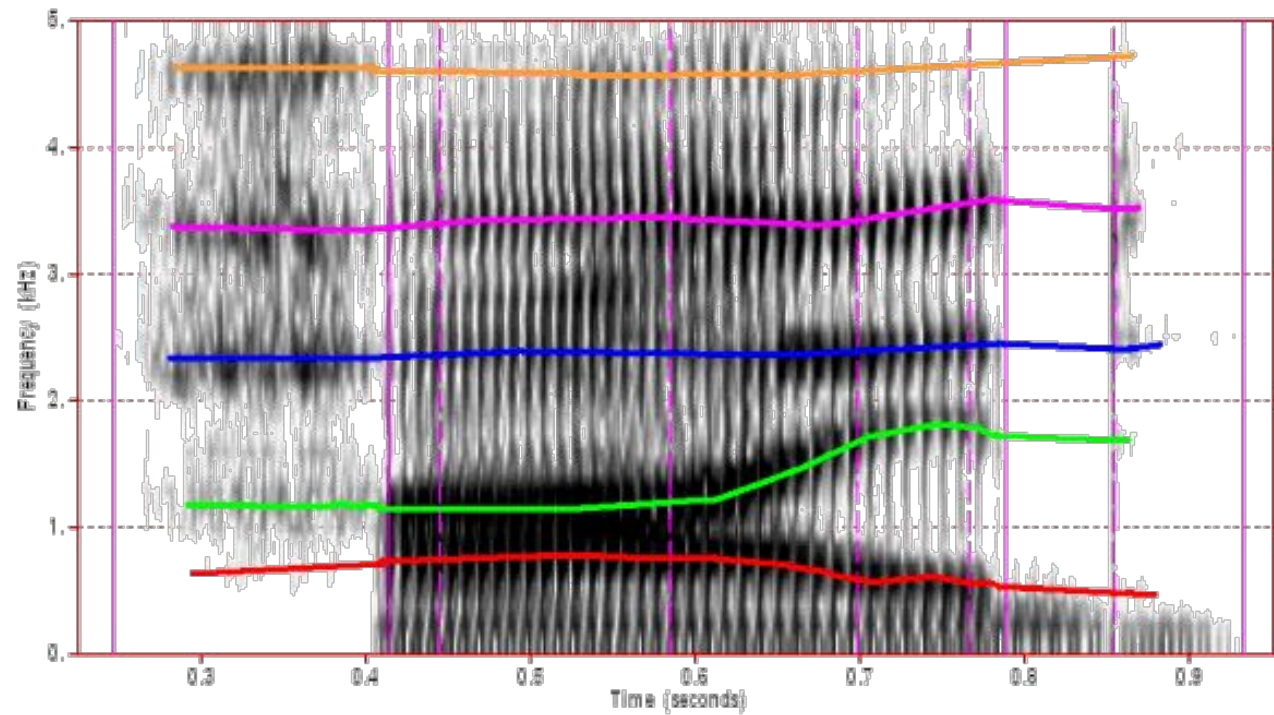


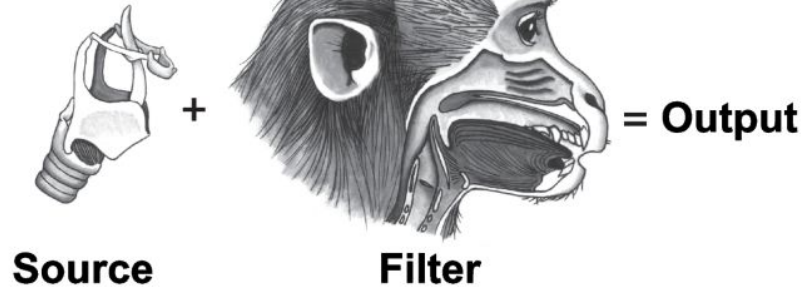
Narrowband Spectrogram - 32 ms Window



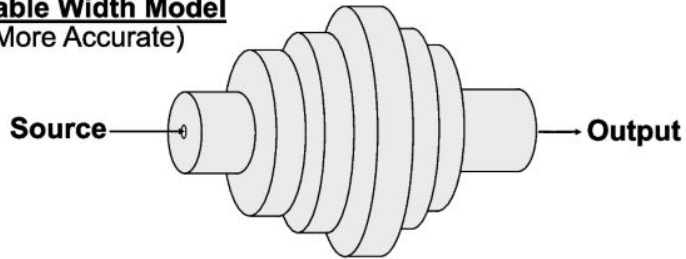
Wideband Spectrogram - 4 ms Window



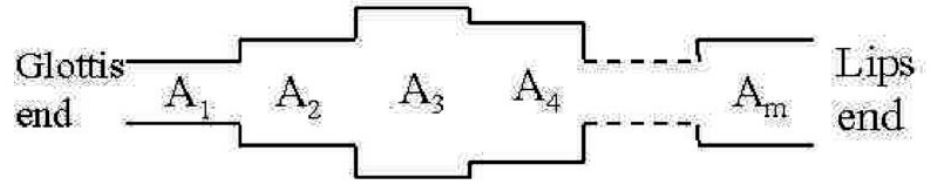
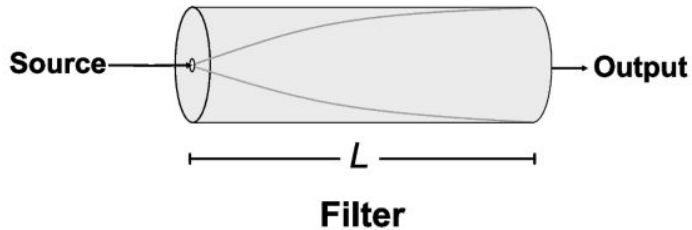


A**B**

Variable Width Model
(More Accurate)



Uniform Tube Model
(Rough Approximation)



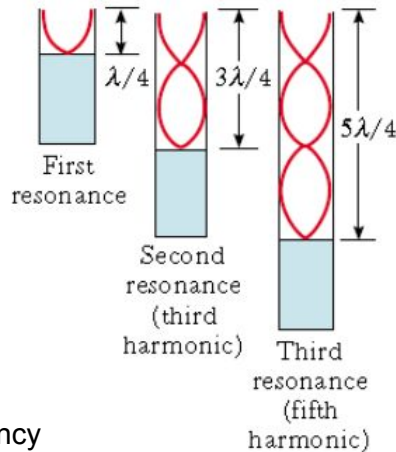
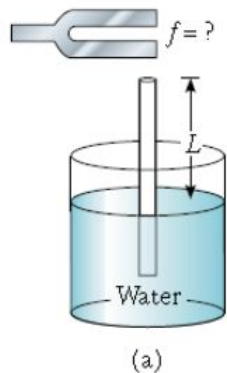
Resonance

At specific frequencies:

- Reflected wave returns in phase (constructive)
- Energy builds up
- Amplitude increases

<https://www.youtube.com/watch?v=xzQO5M1rjfg>

Resonances in uniform tube



(b)

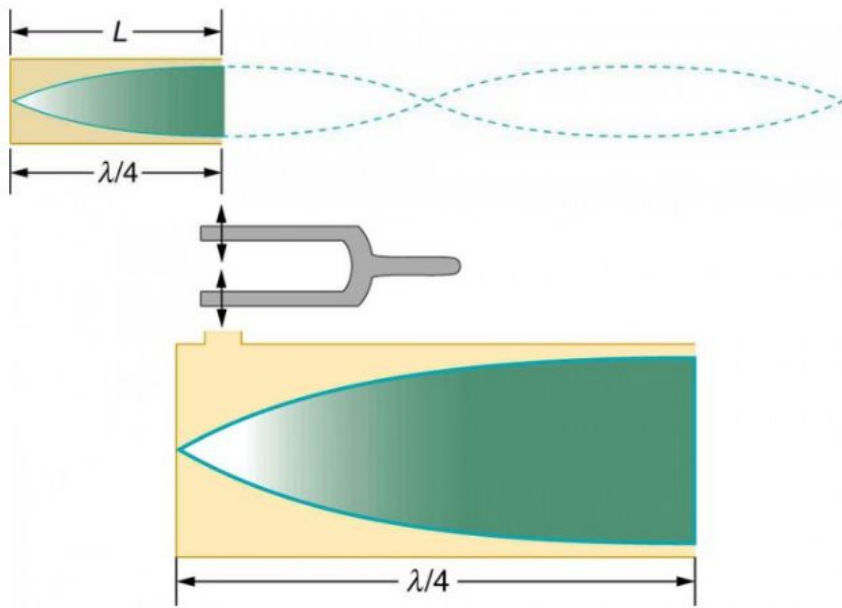
Allowed wavelength & frequency

$$\lambda_n = \frac{4L}{(2n-1)}$$

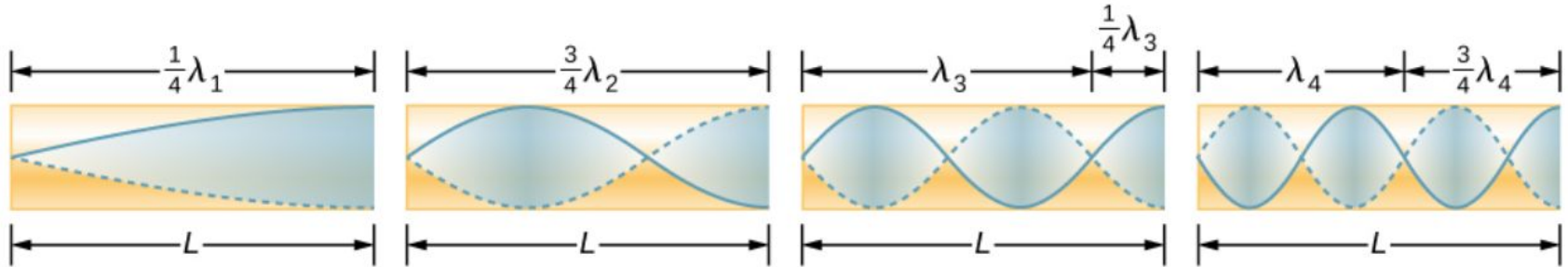
c : velocity of sound

$$f_n = \frac{(2n-1)c}{4L}$$

- The air column has only certain natural frequencies
- If the tuning fork has the right frequency, the air column in the tube resonates loudly
- A standing wave pattern - maximum air displacement at open end & no air displacement at closed end.
- The tube acts like an acoustic filter - the frequency response of which depends on its dimensions



- Length of VT - 17 cm
- Speed of sound - $v=340$ m/sec
- Resonances of the tube: $\frac{nv}{4L}$
500, 1500, 2500 Hz, ...



Fundamental $\lambda_1 = 4L$
 $f_1 = \frac{v}{4L}$

First overtone $\lambda_2 = \frac{4}{3}L$
 $f_3 = \frac{3v}{4L}$

Second overtone $\lambda_3 = \frac{4}{5}L$
 $f_5 = \frac{5v}{4L}$

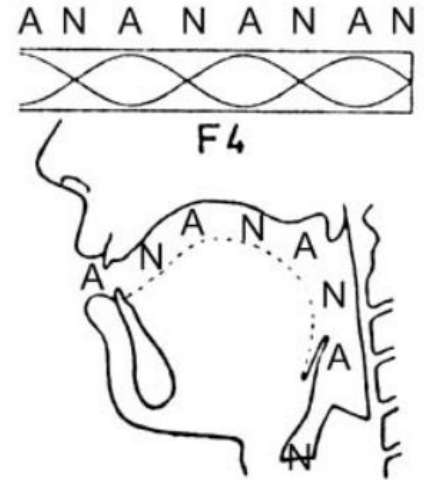
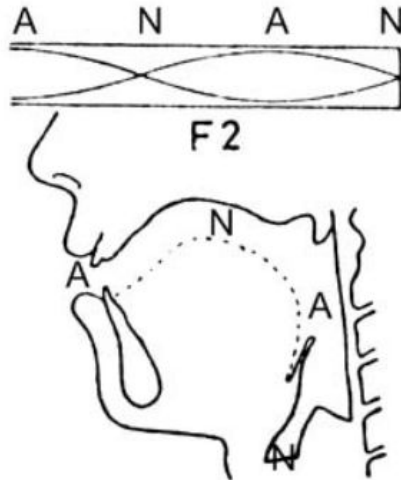
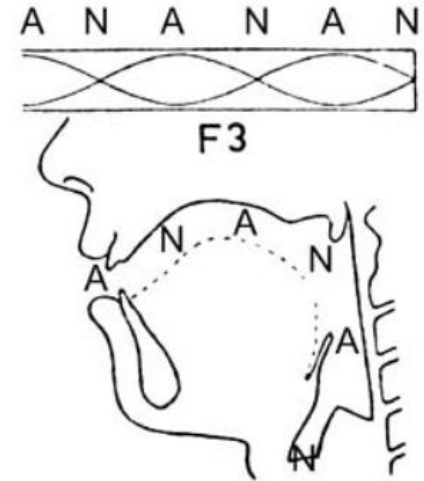
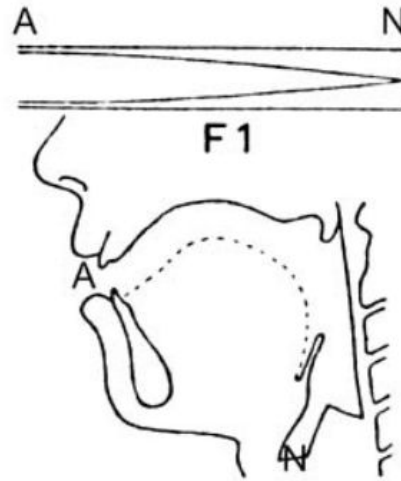
Third overtone $\lambda_4 = \frac{4}{7}L$
 $f_7 = \frac{7v}{4L}$

Wave propagation in VT

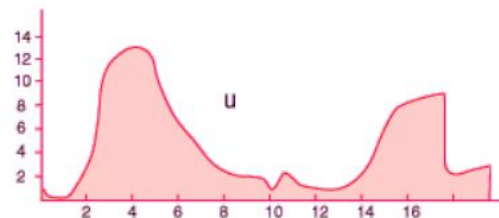
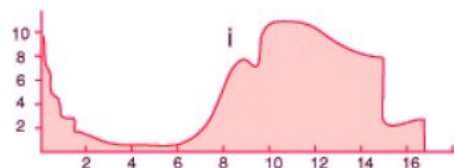
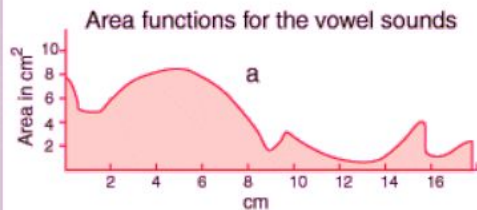
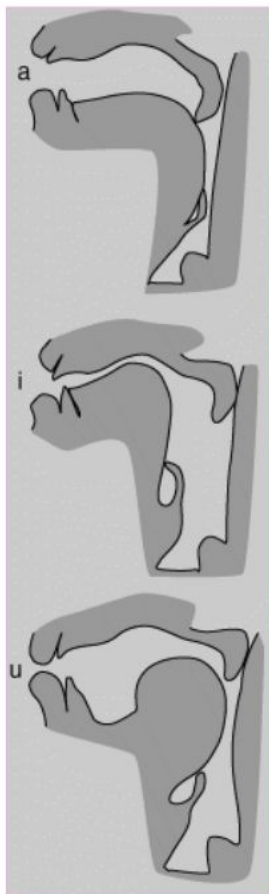
Single tube model

Formant/Resonance frequency: F_1

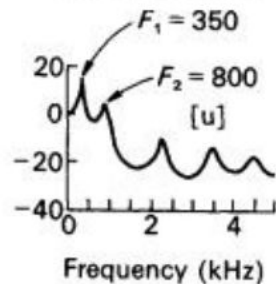
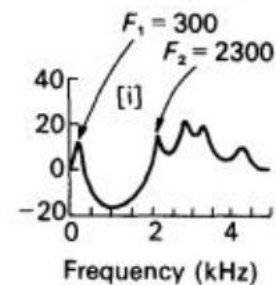
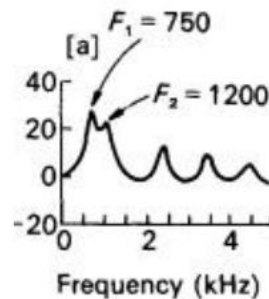
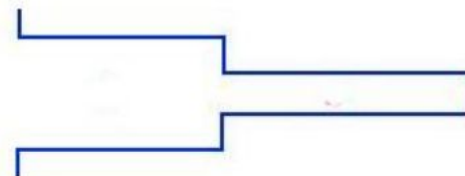
Harmonics: $F_2, F_3, F_4, \dots$

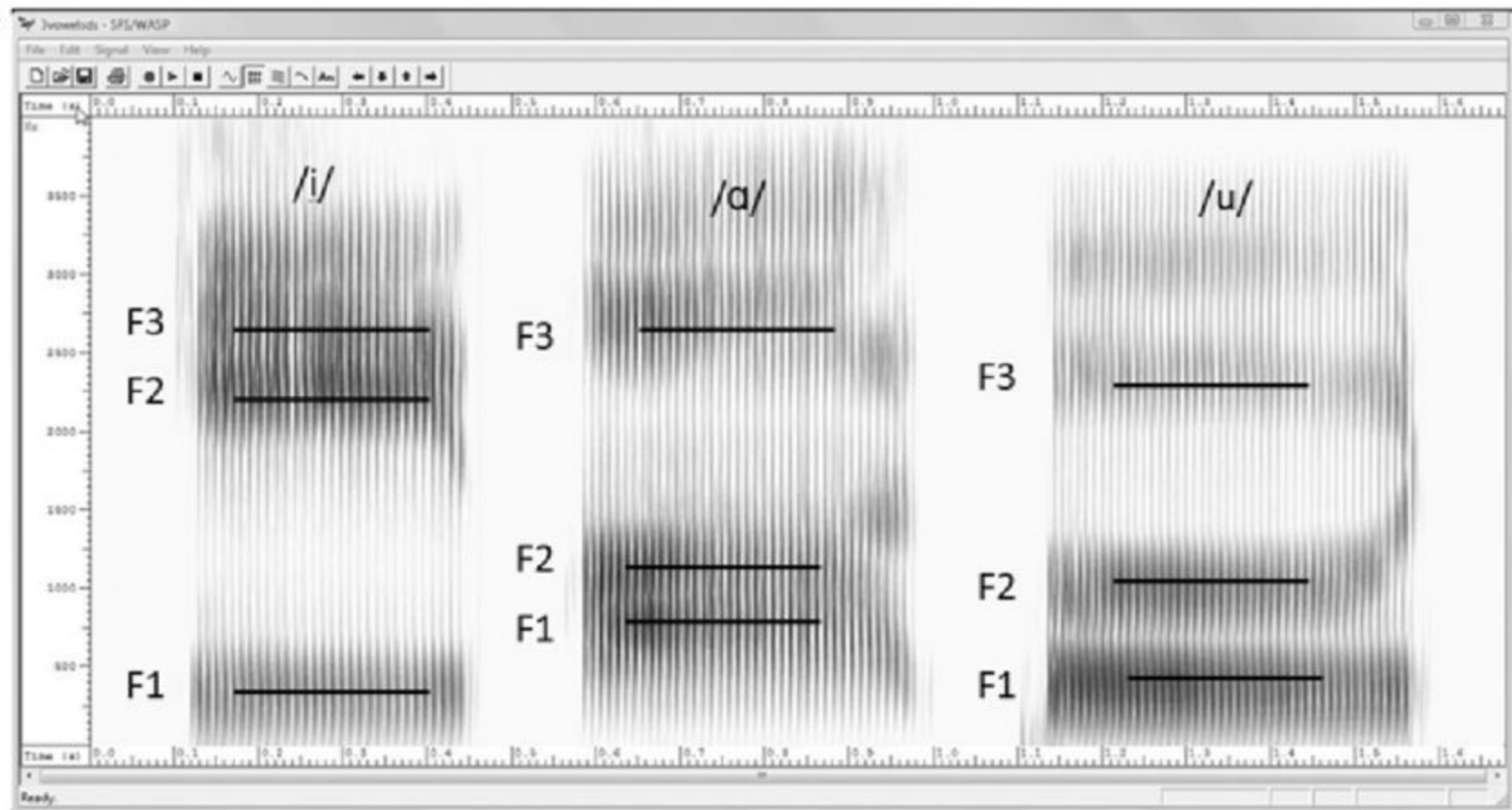


Two tube model

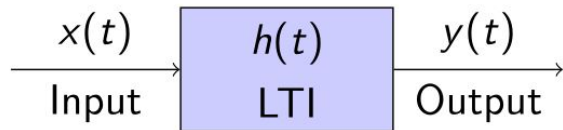


Adapted from Fant





LTI system



- Impulse response: $h(t)$ Frequency response: $H(j\Omega)$
- Output of an LTI system to an arbitrary input $x(t)$

$$y(t) = h(t) * x(t) \longleftrightarrow Y(j\Omega) = H(j\Omega)X(j\Omega)$$

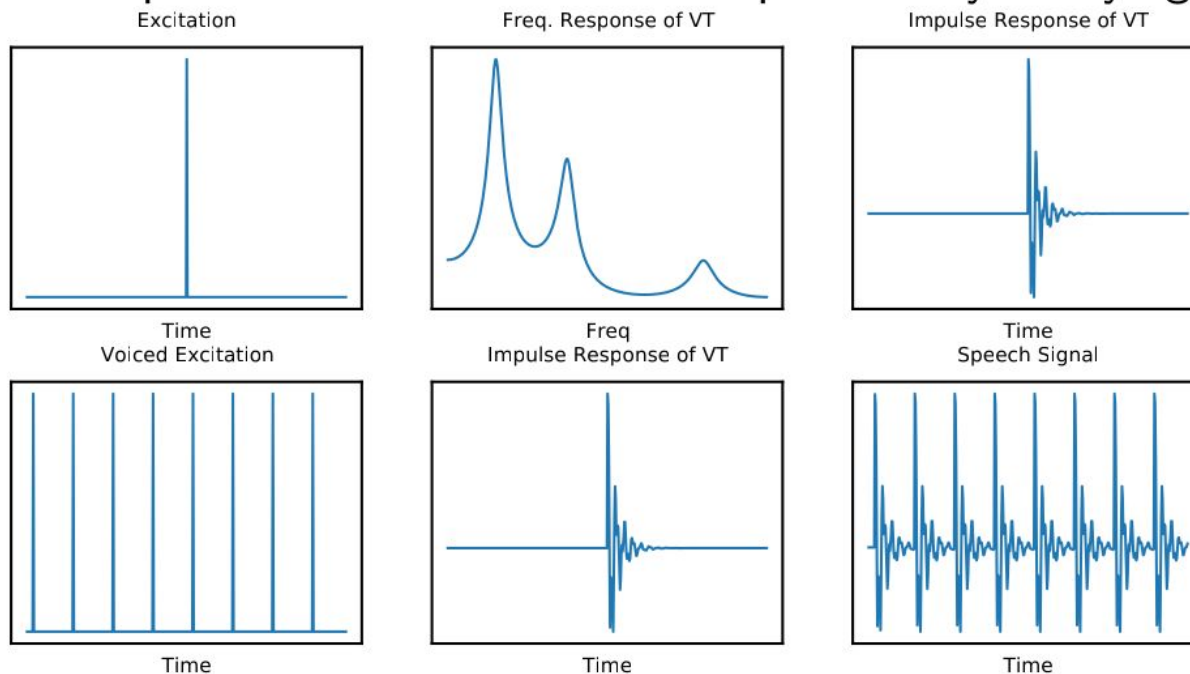
- Output of LTI system to periodic sequence of impulses

$$\sum_{n=-\infty}^{\infty} \delta(t - nT_0) \longleftrightarrow \Omega_0 \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_0) \quad \text{where } \Omega_0 = \frac{2\pi}{T_0}$$

$$y(t) = \sum_{n=-\infty}^{\infty} h(t - nT_0) \longleftrightarrow Y(j\Omega) = \Omega_0 \sum_{k=-\infty}^{\infty} H(jk\Omega_0) \delta(\Omega - k\Omega_0)$$

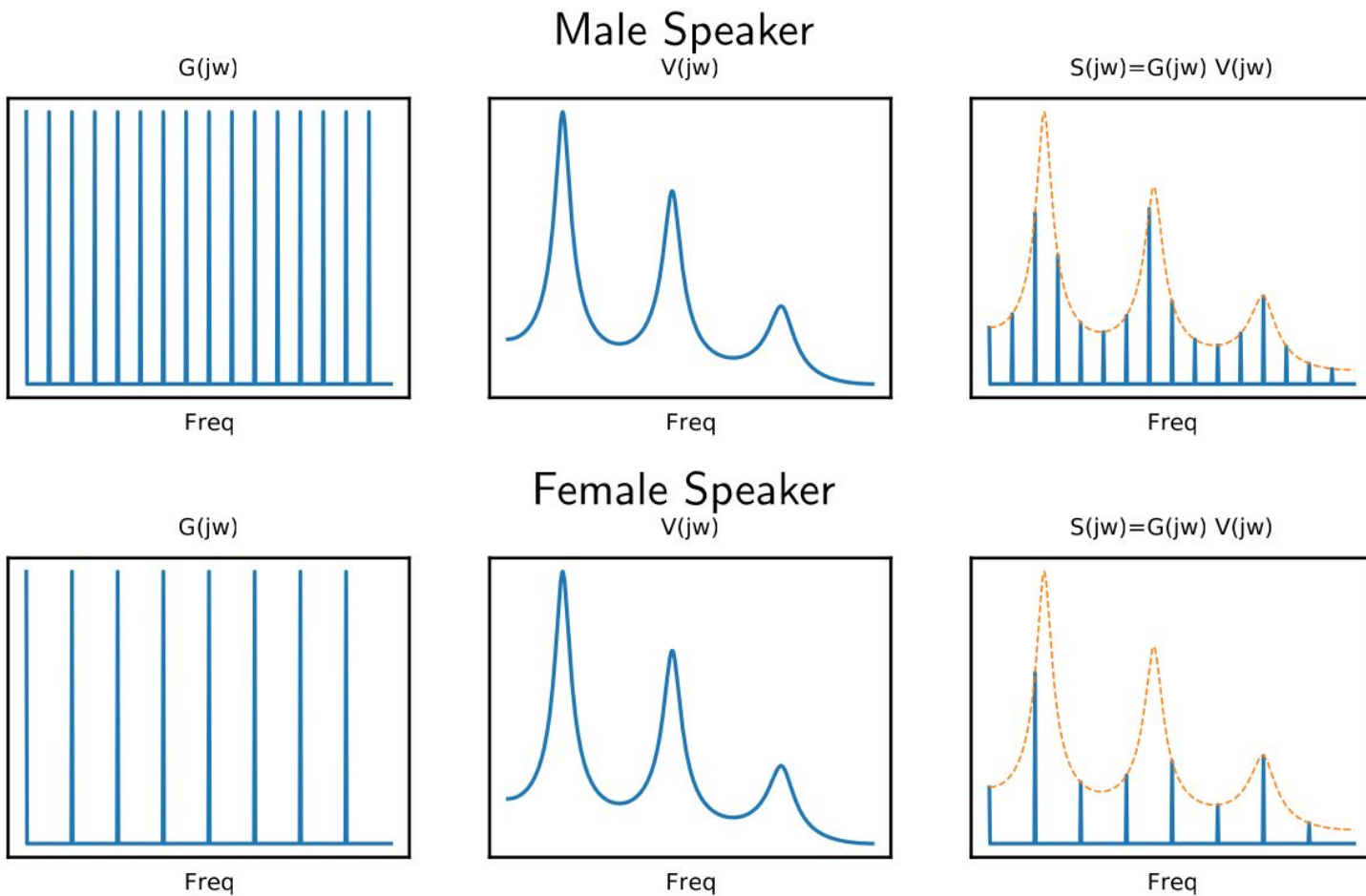
Vocal tract response

- Acoustic tube resonates at specific frequencies (shape & size)
- Resonances of acoustic tube are manifested as spectral peaks
- Impulse response is a combination of exponentially decaying sinusoids

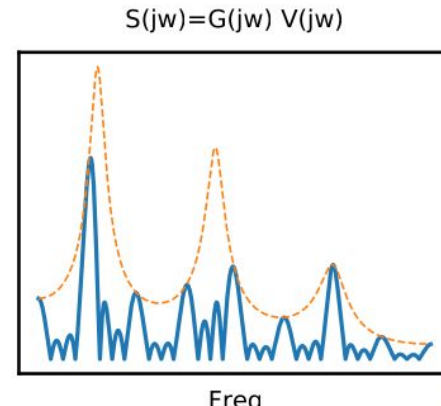
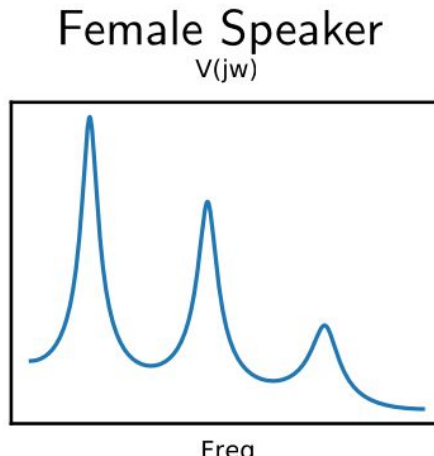
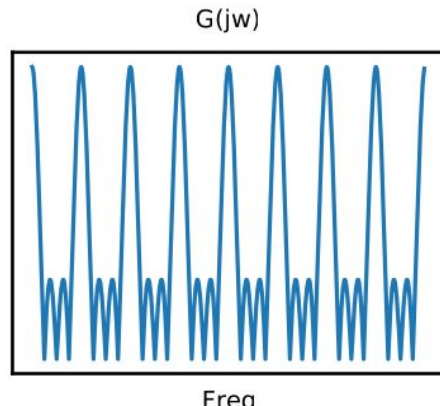
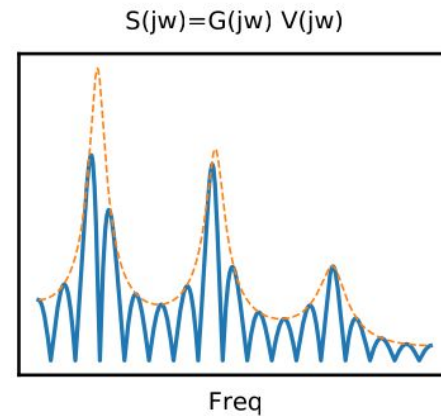
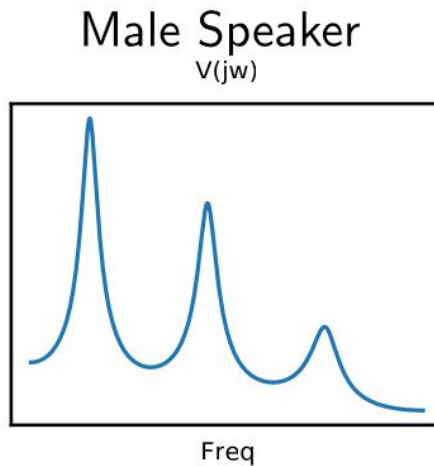
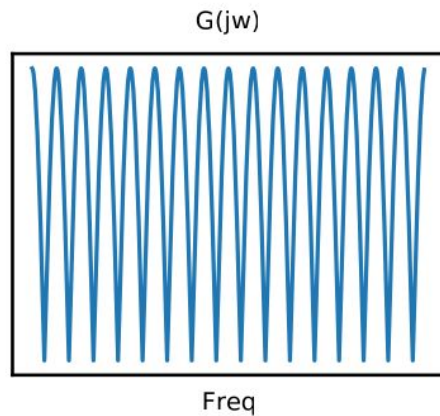


Convolution:

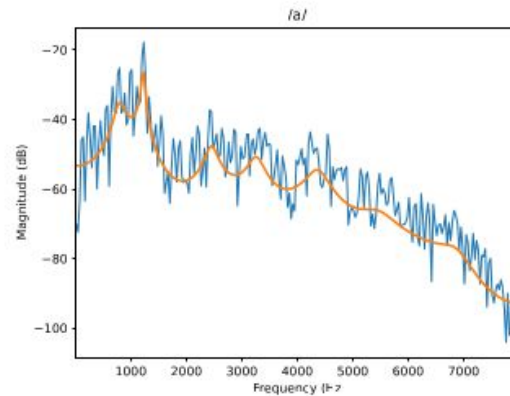
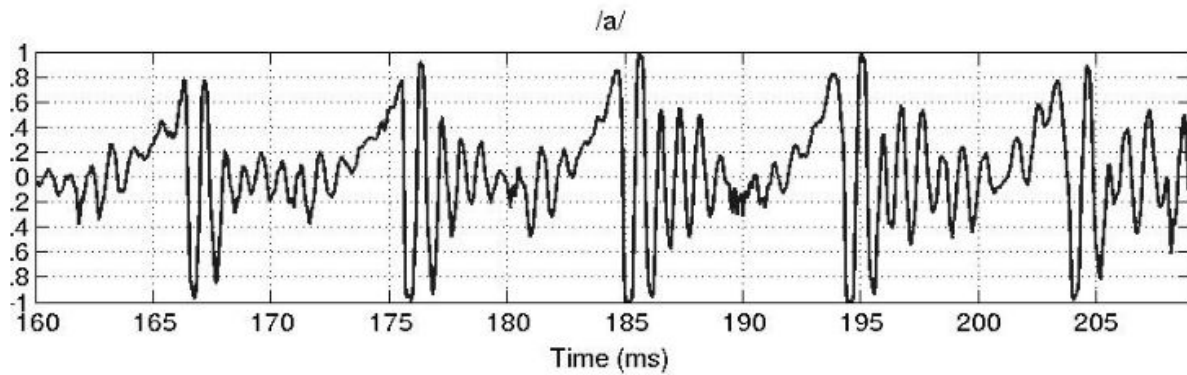
Vowel production



Effect of windowing



Practical observation

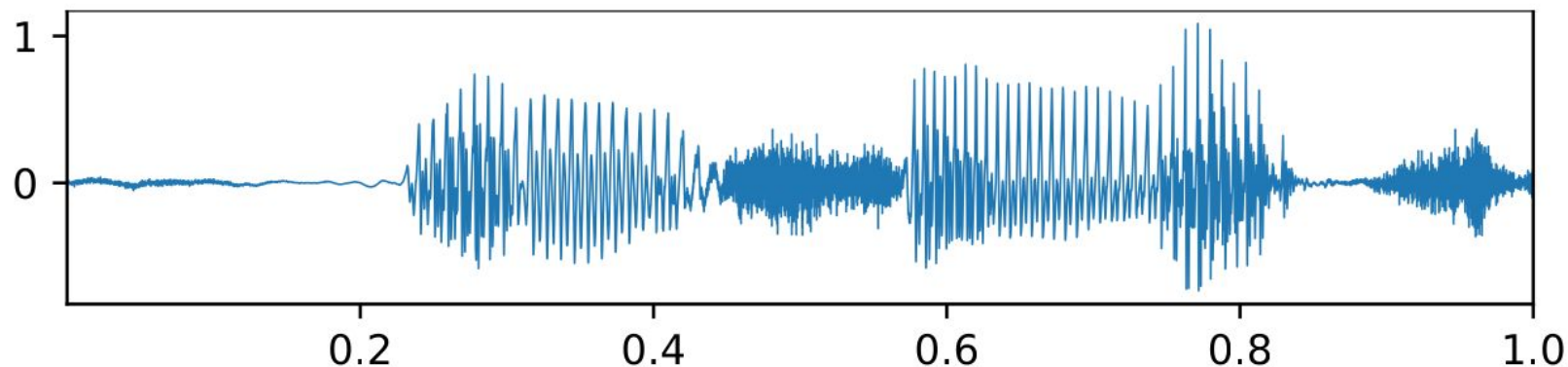




Vowel production summary

- Vowels are produced with a relatively stationary and open VT
- Vowels are produced with voiced excitation
- VT exhibits distinct resonant frequencies - size and shape
- Resonances of the VT are referred to as formant frequencies
- Resonances of VT are manifested as spectral peaks in FD
- It is challenging to estimate formants from speech spectrum
 - Spectral sampling of VT response by periodic impulse-like excitation
 - Inferior performance of speech systems for female speakers
 - Pitch synchronous analysis helps in accurate analysis of VT
- Children have a shorter VT - slightly higher first formant
 - Vocal tract length normalization is used to improve performance

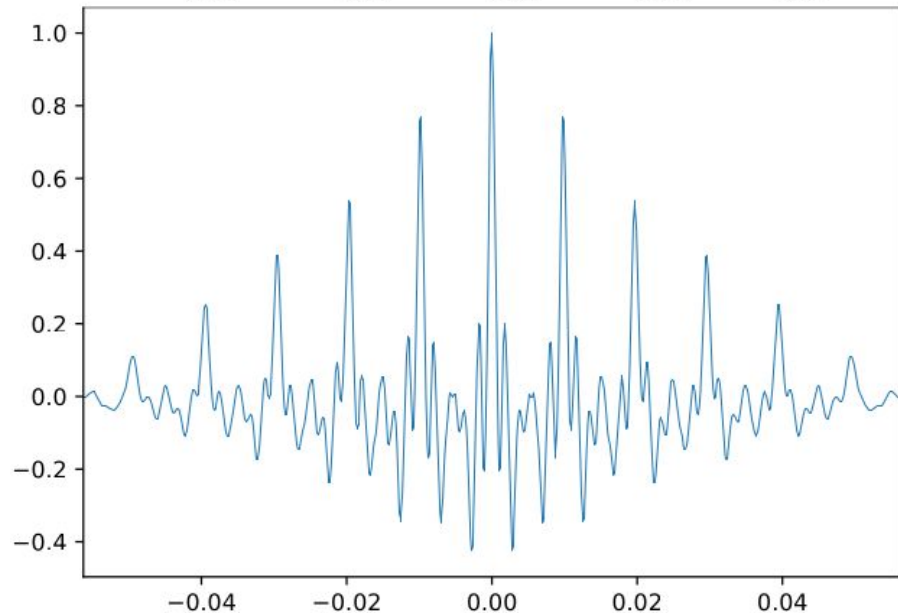
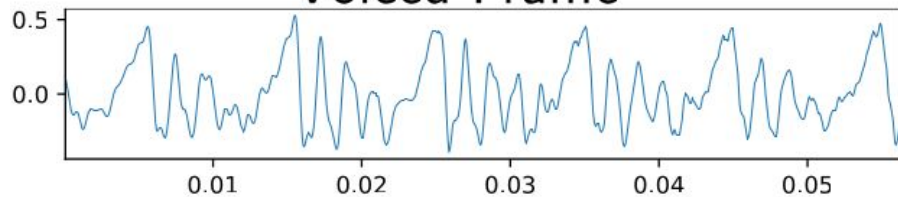
Voice activity detection



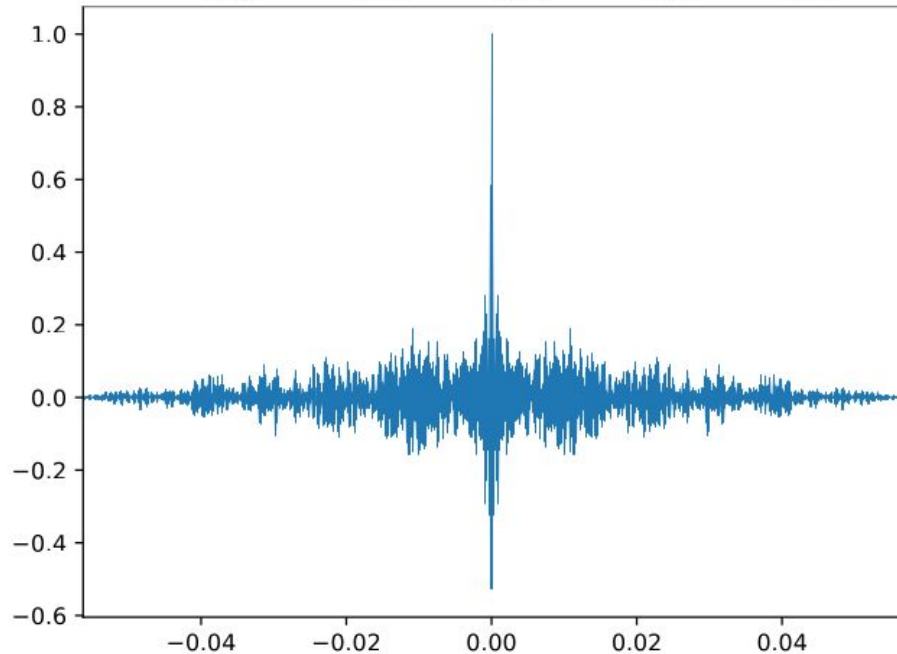
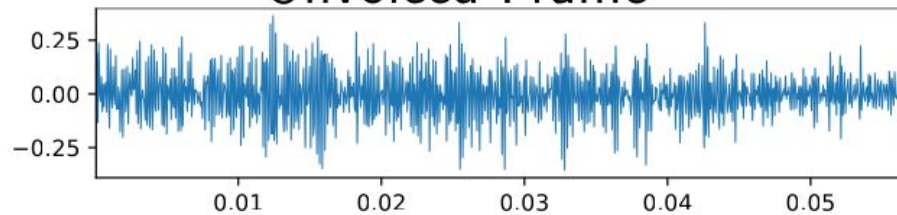
- Identify voiced and unvoiced regions from waveform
- Estimate fundamental frequency in the voiced regions
- Autocorrelation sequence of a mean-subtracted frame $x[n]$

$$r_{xx}[k] = \sum_{n=0}^N x[n]x[n+k] \quad k = \dots, -2, -1, 0, 1, 2, \dots$$

Voiced Frame



Unvoiced Frame

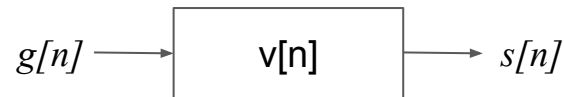


- Strength of the second peak can be used to determine voicing
- Location of the second peak can be used to determine T_0

Voice source (excitation signal)

- Signal analysis:
 - Voiced, unvoiced and silence detection
 - F0 estimation
 - Extraction of glottal closure instants (GCIs)
 - Identifying closed-phase, open-phase and return phase in a laryngeal cycle,
- Gender identification
- Speaker authentication
- Emotion recognition: Call center applications
- Language identification: Rhythm in the phonotactic constraints
- Music analysis: Raga identification, pitch correction
- Voice bank - Preserving and synthesizing voice source

All pole model of vocal tract



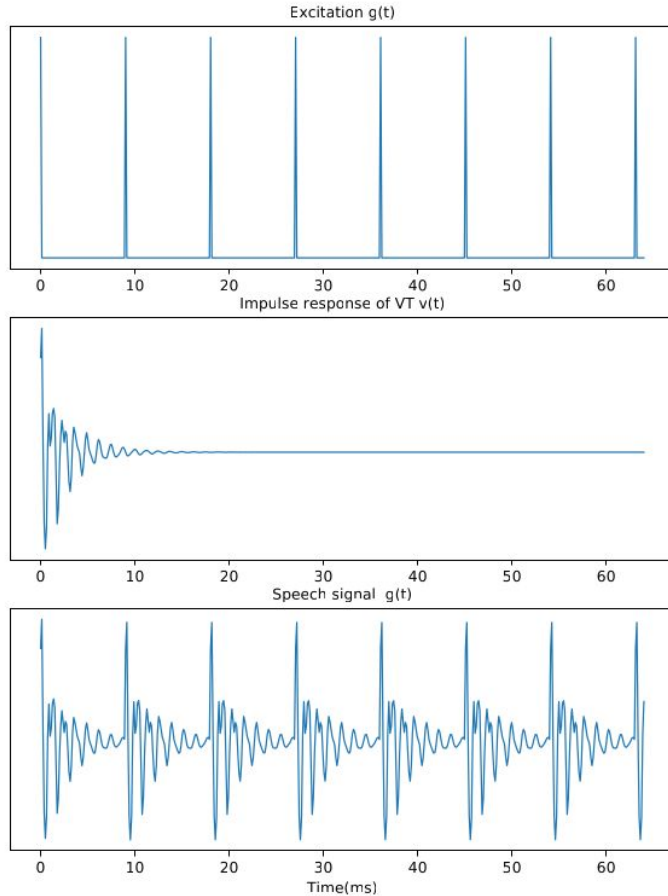
- The resonances of VT show up as spectral peaks
- A spectral peak can be modeled as a complex conjugate pole pair
 $p = re^{j\theta}$ and $p^* = re^{-j\theta}$
 - r corresponds to the bandwidth of the peak
 - θ corresponds to the location of the peak
- VT response with multiple peaks can be approximated as

$$V(z) = \frac{1}{1 + \sum_{k=1}^K a_k z^{-k}}$$

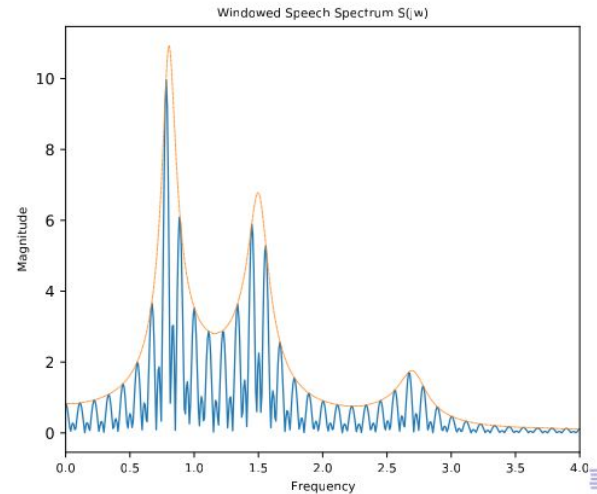
- Since $S(z) = V(z)G(z)$, the speech signal can be modeled as

$$s[n] = - \sum_{k=1}^K a_k s[n-k] + g[n]$$

Generating synthetic signal



- $T_0 = 9ms, F_0 = 111Hz$
- $F_1 = 800, F_2 = 500, F_3 = 2700Hz$
- $r_1 = 0.96, r_2 = 0.94, r_3 = 0.92$
- Sampling frequency $f_s = 8000$ Hz
- Estimate VT characteristics - $s(t)$?



Cepstral analysis of speech: separating source and system

- TD: Convolution

$$s[n] = v[n] * g[n]$$

- FD: Multiplication

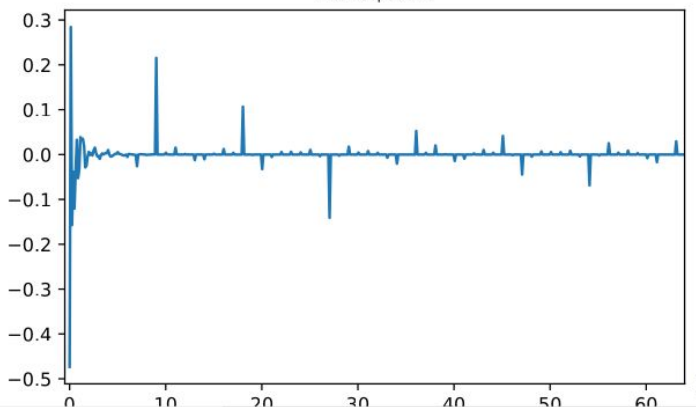
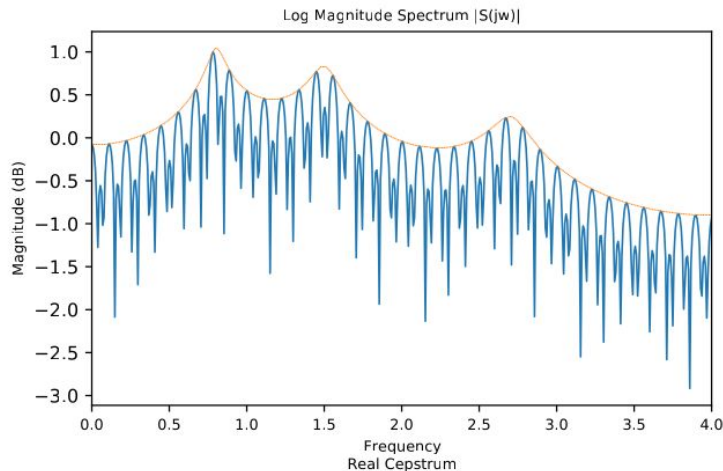
$$S(j\omega) = V(j\omega)G(j\omega)$$

- Log spectral domain: Addition

$$\log S(j\omega) = \log V(j\omega) + \log G(j\omega)$$

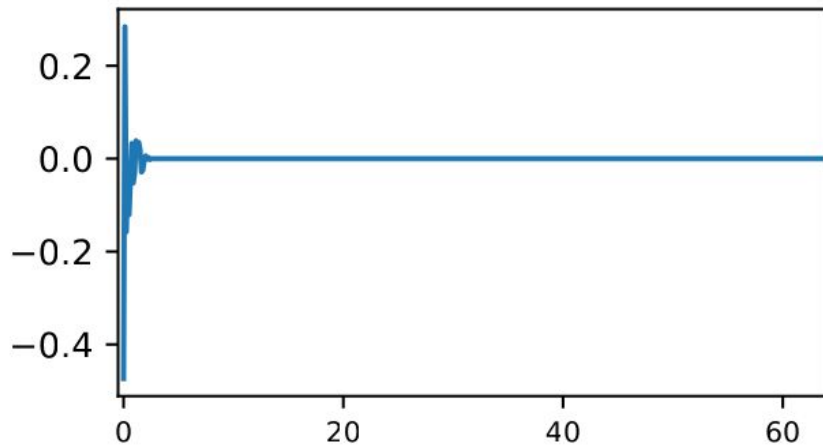
- Inverse FT of log magnitude spectrum is called real cepstrum
- Cepstral domain: Addition

$$s_c[n] = v_c[n] + g_c[n]$$

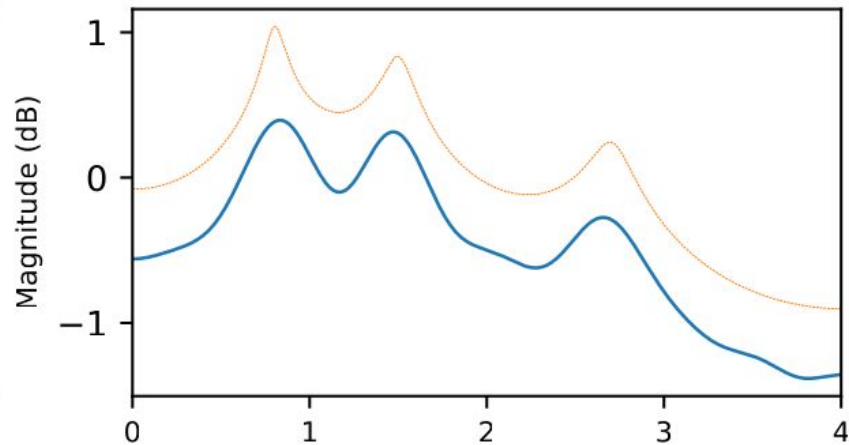


Vocal tract system information

Lowtime Cepstrum

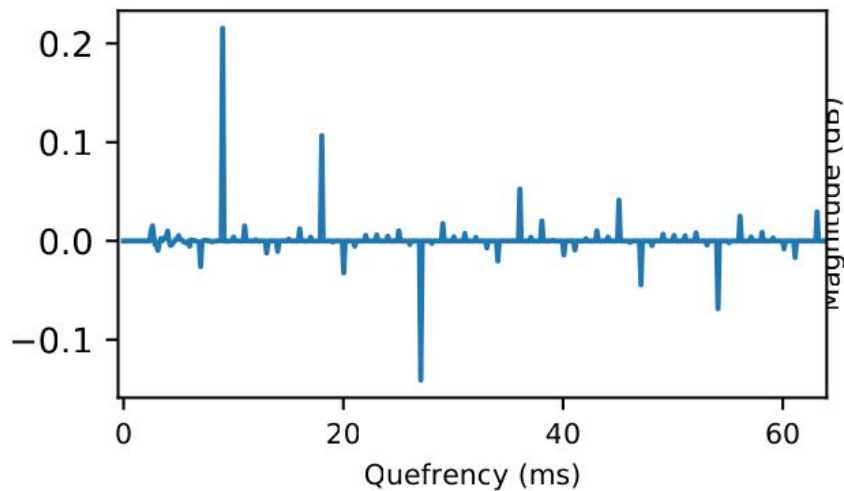


Log Magnitude Spectrum $|S(j\omega)|$

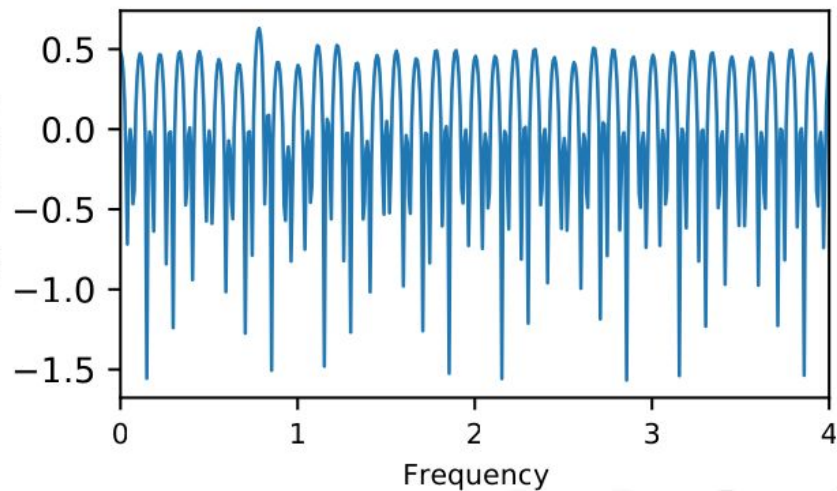


Source information

Hightime Cepstrum



Log Magnitude Spectrum $|S(j\omega)|$

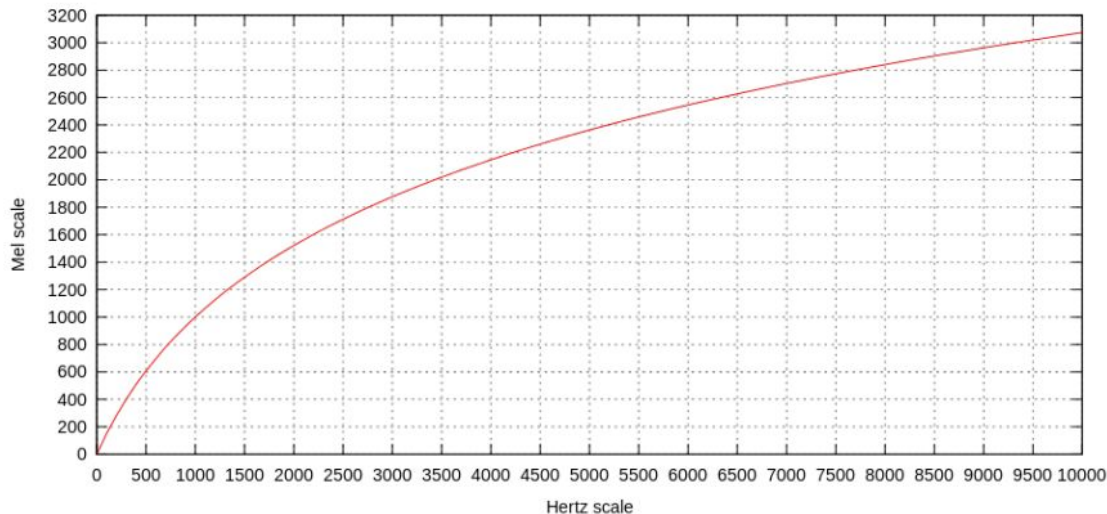


Cepstral extraction

- Algorithm for estimating cepstral features
 - Consider a short window (20-30 ms) of speech signal $s[n]$
 - Discrete Fourier transform: $S[k] = \mathcal{F}\{s[n]\}$
 - Log magnitude spectrum: $\log |S[k]|$
 - Inverse discrete Fourier transform $s_c[n] = \mathcal{F}^{-1}\{\log |S[k]|\}$
 - Retain only first few cepstral coefficients
- Captures gross spectral envelope in a low-dimensional representation
- Higher order is required to resolve two closely spaced spectral peaks
- Gives equal importance to all the frequency components
- Human perception mechanism is sensitive to low frequencies

- Human beings can resolve low-frequency sounds better
- Mel scale incorporates this feature of human perception
- Mel scale is linear up to 1000 Hz, and logarithmic after that
- Filters are placed uniformly-spaced along Mel scale

Mel Scale



$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

Triangular Mel filters

Consider speech signal from 0 to 4KHz

$$\text{mel}(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

$$\text{mel}(0) = 0$$

$$\begin{aligned} \text{mel}(4000) &= 2595 \log(1 + 4000/700) \\ &= 2146 \end{aligned}$$

Considering 20 filters, we need 22 points (21 intervals)

$$\begin{aligned} \text{Mel resolution} &= 2146/21 \\ &= 102.19 \end{aligned}$$

$$\text{ith mel point } m_i = i * 102.19$$

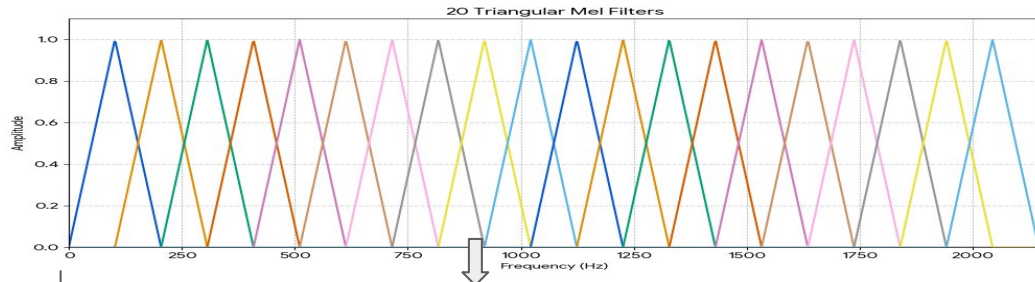
Converting into Hz

$$f_i = 700 \left(10^{m_i/2595} - 1 \right)$$

[0, 102.19, 204.38, 306.57, 408.76, 510.95, 613.14, 715.33, 817.52, 919.71, 1021.90, 1124.09, 1226.28, 1328.47, 1430.66, 1532.85, 1635.04, 1737.23, 1839.42, 1941.61, 2043.80, 2146]

m-th triangular filter

$$H_m(k) = \begin{cases} 0 & k < f(m-1) \\ \frac{k - f(m-1)}{f(m) - f(m-1)} & f(m-1) \leq k \leq f(m) \\ \frac{f(m+1) - k}{f(m+1) - f(m)} & f(m) \leq k \leq f(m+1) \\ 0 & k > f(m+1) \end{cases}$$



$f(m-1)$	$f(m)$	$f(m+1)$
Filter 1: Lower = 0,	Center = 102.19,	Upper = 204.38
Filter 2: Lower = 102.19,	Center = 204.38,	Upper = 306.57
...		
...		
Filter 20: Lower = 1941.61,	Center = 2043.80,	Upper = 2146

m-th filter values are obtained with $k = \text{int} (N * f(m) / f_s)$

Let $N=512$ and $f_s=8000$ then

$k=f(m)=[0, 4, 8, 13, 18, 24, 30, 36, 44, 51, 60, 69, 79, 89, 101, 114, 128, 143, 160, 179, 200, 222, 256]$

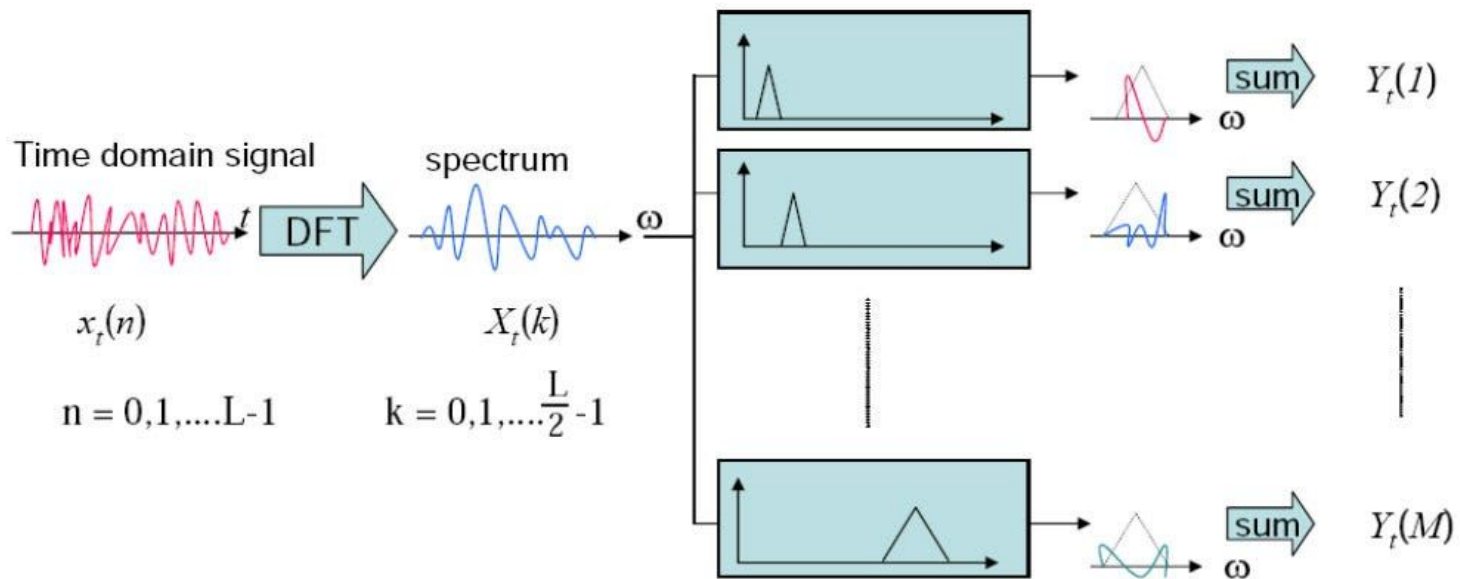
First filter:

$$f(0)=0$$

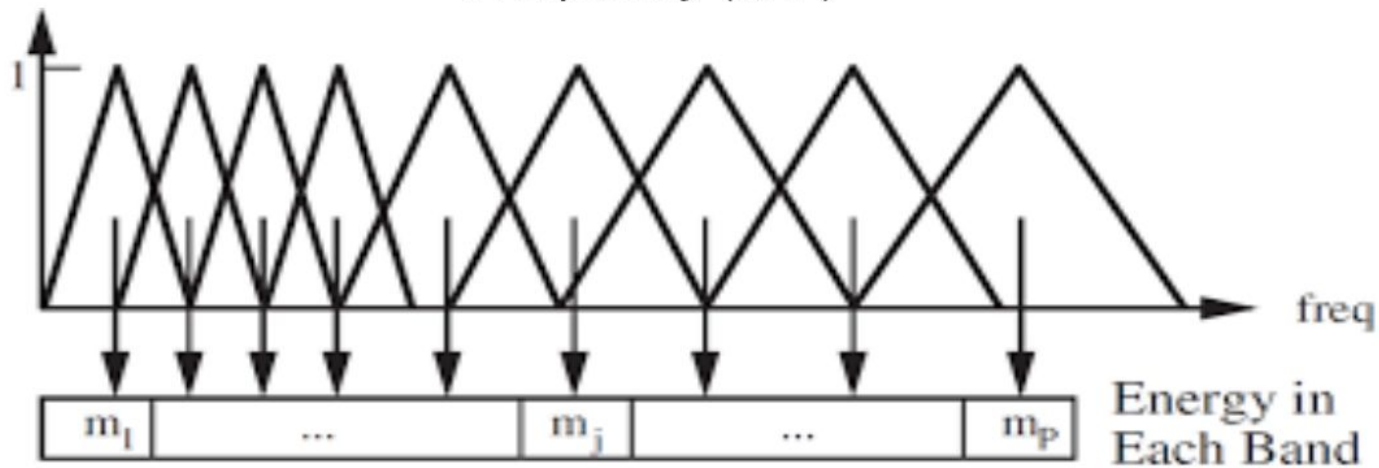
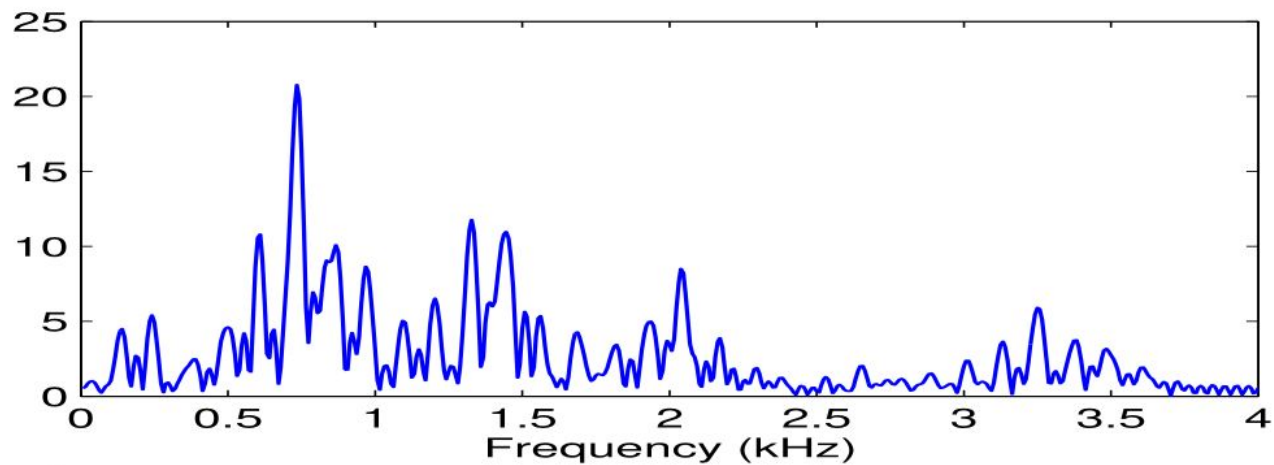
$$f(1)=4$$

$$f(2)=8$$

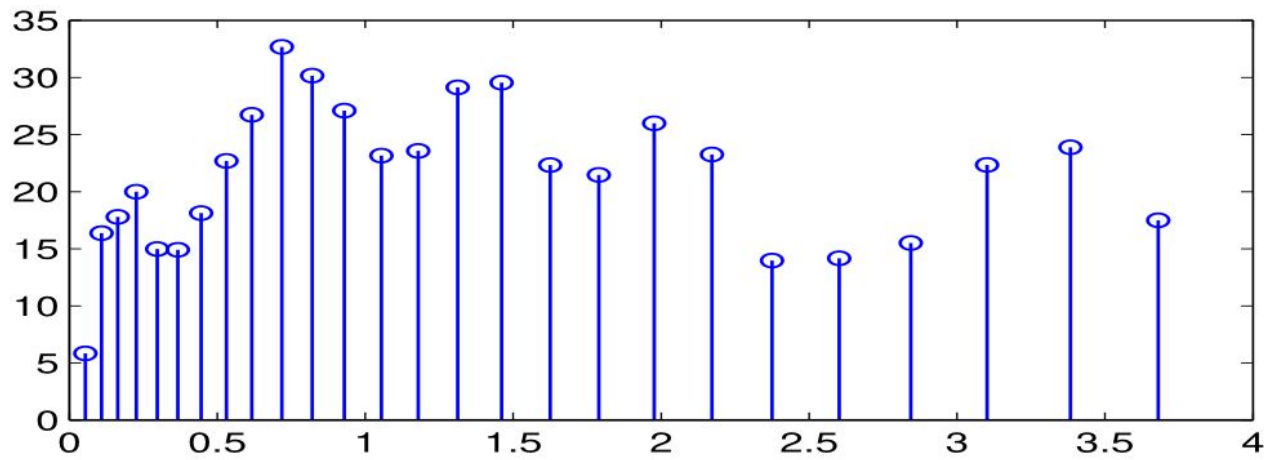
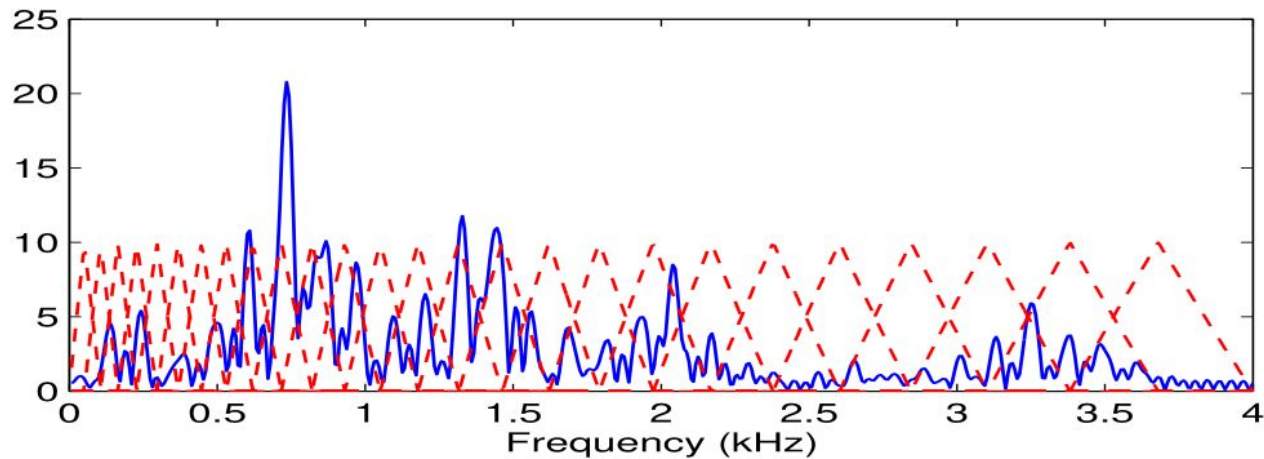
$$H_1(k) = \begin{cases} 0 & k < 0 \\ \frac{k}{4} & 0 \leq k \leq 4 \\ \frac{8-k}{4} & 4 \leq k \leq 8 \\ 0 & k > 8 \end{cases}$$

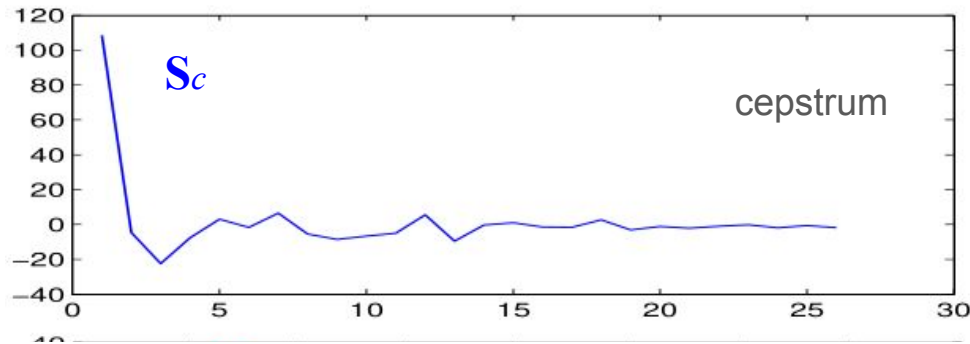
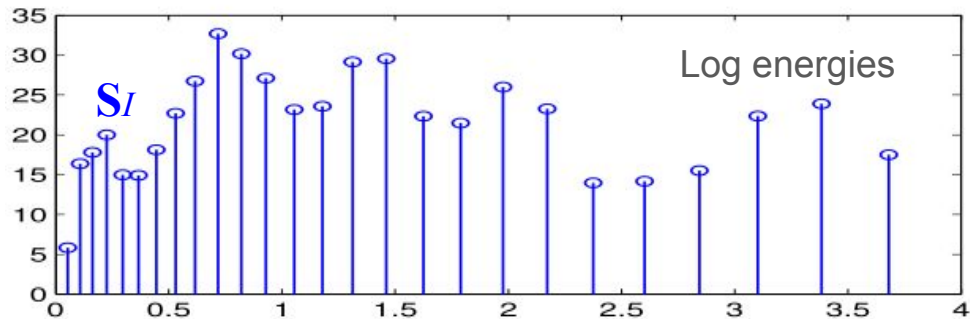
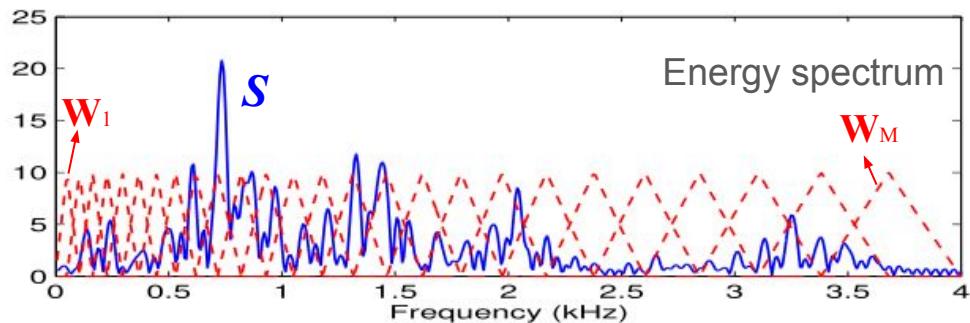


Mel filters



Mel filterbank energy coefficients





MFCCs from waveform

- 20-30 ms window: $s[n]$
- N-point DFT - $S[k]$
- Squared magnitude $|S[k]|^2$
- S : symmetric half of $|S[k]|^2$
- Energy over Mel-filter bands

$$\mathbf{S}_m = \mathbf{W}_{M \times \frac{N}{2}} \mathbf{S}_{\frac{N}{2} \times 1}$$

$\mathbf{W}$ is the Mel-weight matrix

- Log energy - $\mathbf{S}_l = \log \mathbf{S}_m$
- Compute IDCT: $\mathbf{S}_c = \mathbf{D}_{M \times M} \mathbf{S}_l$
- Retain first few coefficients $\hat{\mathbf{S}}_c$